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Randomized Numerical Linear Algebra in Mixed Precision, with an Application to Climate Models

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Contents

Introduction

This thesis studies randomized numerical linear algebra and mixed-precision arithmetic with emphasis on covariance-based dependence correction. The motivation is simple. Many linear algebra problems arising in scientific computing are large enough that classical dense methods become expensive, and modern hardware no longer offers one uniform arithmetic. Instead, several precisions are available, often with very different speed, memory cost, and numerical reliability.

Randomized numerical linear algebra addresses the first difficulty. The basic idea is to replace expensive deterministic operations by randomized approximations that preserve the important geometry of the problem. Typical examples are sketching, random projections, and structured random transforms. These methods are especially natural for overdetermined least-squares problems, where the cost is dominated by the size of the data matrix rather than by the dimension of the solution space.

Mixed-precision arithmetic addresses the second difficulty. Instead of performing the whole computation in one format, one tries to use lower precision where it is cheap and higher precision where it is needed for stability or accuracy. This may reduce runtime and memory use, but it also changes the perturbation model of the algorithm. A method that is reliable in one precision may behave quite differently in another.

In this thesis, we are interested in the interaction of these two ideas. Sketching changes the problem by replacing it with a compressed surrogate. Low precision changes the arithmetic by introducing larger rounding errors. Both effects act on the same operator-theoretic problem, and both are filtered through conditioning. Therefore, they should not be studied separately.

The central mathematical contribution pursued here is a mixed-precision randomized low-rank covariance transport map for multivariate dependence correction. Least squares remains essential, but now as the first model problem in which sketching, conditioning, and finite precision can be analyzed in a clean setting. We therefore begin with randomized least-squares solvers based on Gaussian sketches and structured transforms such as the subsampled randomized Hadamard transform, and then transfer the same perturbation viewpoint to regularized covariance operators and the matrix functions used in dependence correction.

The climate-model material appears later as an application rather than as the main topic of the thesis. Its role is to test whether methods developed on synthetic and theoretical examples remain useful in a more realistic workflow. The outcome need not be a pure speedup. Reduced memory use, acceptable lower-precision behavior, or a clearer picture of failure regimes are also valuable results.

Main Contribution

The thesis advances three interlocking contributions.

1. **Perturbation analysis for regularized covariance transport.** We give deterministic first-order bounds for the perturbation of the regularized transport map $T_\lambda = (C_X + \lambda I)^{-1/2}(C_Y + \lambda I)^{1/2}$ under additive disturbances to the source and target covariance operators. The bounds make explicit how the inverse-square-root amplification factor $K_\lambda(C_X, C_Y)$ depends on the regularized spectral gap, and they apply whether the perturbations arise from randomized low-rank compression, floating-point arithmetic, or both.

2. **Randomized low-rank approximation and mixed-precision error decomposition.** We decompose the total covariance perturbation into a randomized truncation term governed by the spectral tail beyond the chosen rank, and a floating-point term governed by the unit roundoff, the accumulation length, and the norm of the sketching operator. This decomposition is the natural quantitative setting for the balancing-inexactness principle: mixed precision is safe only while $\|E_{\mathrm{fp}}\| \lesssim \|E_{\mathrm{rand}}\|$, and the regime boundary is computable from the rank, the oversampling, and the spectrum of the covariance operator.

3. **Synthetic and climate-shaped experiments identifying rank, regularization, and precision regimes.** We design experiments that vary the approximation rank, the regularization level λ, and the working precision in the sketch products, while tracking covariance error, transport-map error, and the induced dependence error on corrected future data. The experiments distinguish three regimes: an approximation-dominated regime in which lower precision is hidden beneath truncation error, a regularization-dominated regime in which transport instability is the bottleneck, and a precision-visible regime in which floating-point error overtakes the randomized approximation. Negative or inconclusive randomized results are framed as controlled regime-identification findings rather than as failed speedup attempts.

Taken together, these contributions turn the experimental observation that randomized covariance transport is numerically viable only in certain parameter regimes into a coherent thesis-level claim: the regime boundary is predictable from the spectral structure of the covariance operators and the regularization parameter, and mixed precision is safe only inside the approximation-dominated sub-regime.

The structure of the thesis is as follows. The preliminary chapters fix notation and collect the basic mathematical material. The next chapters discuss mixed precision and randomized least squares in more detail, first as model problems and then as technical tools. They are followed by a chapter on randomized covariance transport, where the main dependence-correction map is formulated and its perturbation structure is analyzed. The numerical experiments then test both the least-squares and covariance-transport viewpoints. A short atmospheric context chapter explains how the application variables and dependence structures arise in climate-model data, and the final substantive chapter returns to the climate-model application.

Notation & Conventions

Throughout the thesis, uppercase Latin letters such as A, S, and Q denote matrices. At the same time, they are often interpreted as concrete representations of linear maps between finite-dimensional inner-product spaces. Vectors are denoted by lowercase letters such as x and b. For a least- squares problem we usually write $A \in \mathbb{R}^{n\times d}$ with $n \geq d$, and we consider the overdetermined problem

$$\min_{x\in\mathbb{R}^d} \|Ax - b\|_2.$$

The symbol S is reserved for a sketching matrix, or equivalently for a linear map that compresses the ambient data space. A sketched least-squares problem therefore has the form

$$\min_{x\in\mathbb{R}^d} \|SAx - Sb\|_2.$$

Unless stated otherwise, $\|\cdot\|_2$ denotes the Euclidean norm for vectors and the induced operator norm for matrices, while $\|\cdot\|_F$ denotes the Frobenius norm. The condition number of a nonsingular matrix M in the Euclidean norm is written as

$$\kappa_2(M) = \|M\|_2\|M^{-1}\|_2$$

and is interpreted as the sensitivity of the associated linear system to perturbations.

Floating-point quantities are discussed using the standard model of rounding to nearest. The unit roundoff of a format is denoted by u. When mixed precision is used, different unit roundoffs may appear in different parts of the same algorithm. These are distinguished by subscripts such as u_ℓ for lower precision and u_h for higher precision.

In numerical experiments, the terms “solution error” and “residual error” are used in the following sense:

- solution error compares an approximation $\hat{x}$ directly with a reference solution $x_\star$, typically through $\|\hat{x} - x_\star\|_2$ or a relative variant,
- residual error measures how well $\hat{x}$ approximates the data vector b through quantities such as $\|A\hat{x} - b\|_2$.

Gaussian sketches, structured sketches, and mixed-precision variants are defined more carefully in the later chapters. The notation fixed here remains standard throughout the text. We also move freely between matrix language and the corresponding operator viewpoint whenever this makes the numerical question clearer.

1 Mathematical Background

This chapter collects the mathematical material needed later for randomized least-squares methods and for covariance-based dependence correction. The main ingredients come from four directions: classical least-squares theory, random projections and subspace embeddings, symmetric positive definite operators and their matrix functions, and perturbation questions related to conditioning. Although all spaces in this thesis are finite-dimensional, it is often convenient to use the language of linear maps, ranges, and subspaces rather than only matrix coordinates.

For an overdetermined system $A \in \mathbb{R}^{n\times d}$ with $n \geq d$, the least-squares problem

$$\min_{x\in\mathbb{R}^d} \|Ax - b\|_2$$

asks for the vector whose image under A is closest to b in Euclidean norm. Geometrically, this means projecting b onto the range of A. If A has full column rank, the minimizer is unique. In exact arithmetic, it may be characterized by the normal equations or, more stably, by an orthogonal factorization.

The randomized viewpoint replaces the original problem by a smaller one. A sketching matrix $S \in \mathbb{R}^{s\times n}$, with s much smaller than n, is chosen so that the geometry of the range of A is approximately preserved after compression. One then solves

$$\min_{x\in\mathbb{R}^d} \|SAx - Sb\|_2.$$

The quality of this replacement depends on how well the sketch preserves norms and angles on the relevant subspace. This is where subspace embeddings and the Johnson–Lindenstrauss phenomenon enter the picture. The sketch should act almost isometrically on the subspace that carries the least-squares geometry. The formal definitions and embedding guarantees are given in Chapter 3 →p.13.

Two sketch families are especially important in this thesis. The first is the Gaussian sketch, whose entries are independent random variables with suitable scaling. The second is the class of structured sketches, especially transforms based on randomized Hadamard matrices. Gaussian sketches are the cleanest theoretical baseline. Structured sketches are more attractive computationally because they can exploit fast transforms.

Conditioning plays a central role throughout. Even when sketching gives a good approximation in principle, the numerical behavior of the reduced problem depends strongly on the conditioning of SA, that is, on the stability of the compressed operator on the relevant subspace. This becomes even more important in finite precision, since rounding errors may be harmless for a well-conditioned problem and severe for an ill-conditioned one. The relation between sketching quality and conditioning is discussed in Proposition 3.7 →p.18.

The later mixed-precision discussion therefore rests on a simple principle. Randomized approximation error and floating-point error should not be studied in isolation. A useful theory must account for both at the same time.

1.1 Singular Value Decomposition and Pseudoinverse

The basic structural tool for least-squares analysis is the thin singular value decomposition. If $A \in \mathbb{R}^{n\times d}$ has rank r, then

$$A = U_r \Sigma_r V_r^\top,$$

where $U_r \in \mathbb{R}^{n\times r}$ and $V_r \in \mathbb{R}^{d\times r}$ have orthonormal columns and

$$\Sigma_r = \operatorname{diag}(\sigma_1, \dots, \sigma_r), \qquad \sigma_1 \geq \cdots \geq \sigma_r > 0,$$

contains the nonzero singular values. This factorization separates the geometry of the range, the nullspace, and the spectral scales of the operator.

If A has full column rank, the least-squares minimizer may be written as

$$x_\star = A^\dagger b, \qquad A^\dagger = (A^\top A)^{-1} A^\top = V_r \Sigma_r^{-1} U_r^\top,$$

where $A^\dagger$ is the Moore–Penrose pseudoinverse. This formula is useful for analysis because it makes explicit that the small singular values of A are the source of sensitivity: directions associated with $\sigma_j \ll 1$ are amplified by σ_j^{-1}.

The operator norm of A is

$$\|A\|_2 = \sigma_1,$$

and for a full-column-rank matrix the Euclidean condition number is

$$\kappa_2(A) = \frac{\sigma_1}{\sigma_r}.$$

This quantity measures how strongly data perturbations can be amplified by the least-squares solve and reappears later in the analysis of sketched systems.

1.2 Normal Equations, QR, and SVD

The least-squares problem admits several algebraic formulations, but they do not behave equally well numerically. The normal equations

$$A^\top A x = A^\top b$$

are attractive because they reduce the problem to a square system. Their drawback is that they square the spectrum:

$$\sigma_j(A^\top A) = \sigma_j(A)^2, \qquad \kappa_2(A^\top A) = \kappa_2(A)^2.$$

Thus a moderately ill-conditioned least-squares problem can become a severely ill-conditioned linear system when written in this form.

By contrast, if $A = QR$ is a thin QR factorization with Q having orthonormal columns and $R \in \mathbb{R}^{d\times d}$ upper triangular, then the least-squares problem reduces to

$$\min_x \|Rx - Q^\top b\|_2,$$

or equivalently to the triangular solve

$$Rx = Q^\top b$$

in the full-column-rank case. QR avoids the explicit formation of $A^\top A$ and is therefore the standard stable dense formulation.

The SVD is more expensive than QR but more informative. It reveals rank deficiency directly, yields the pseudoinverse explicitly, and makes clear how solution sensitivity depends on the singular spectrum. In the present thesis, QR is the practical dense baseline, while the SVD is the conceptual reference that makes the conditioning questions transparent [1].

1.3 Least-Squares Perturbation Viewpoint

The numerical sensitivity of least squares depends on more than the matrix norm alone. Let $x_\star$ solve

$$\min_x \|Ax - b\|_2,$$

and write $r_\star = b - Ax_\star$ for the optimal residual. Classical perturbation theory shows that under small perturbations ΔA and Δb, the relative change in the minimizer is controlled by the condition number together with a term that reflects the size of the residual. Schematically,

$$\frac{\|\Delta x_\star\|_2}{\|x_\star\|_2} \lesssim \kappa_2(A)\left(\frac{\|\Delta A\|_2}{\|A\|_2} + \frac{\|\Delta b\|_2}{\|b\|_2}\right) + \kappa_2(A)^2 \frac{\|r_\star\|_2}{\|A\|_2\,\|x_\star\|_2}\,\frac{\|\Delta A\|_2}{\|A\|_2}\,,$$

with more precise constants depending on the chosen formulation Ch. 1 [1]. The important qualitative message is that solution sensitivity is mild when the residual is small and A is well conditioned, but can be much worse in inconsistent or ill-conditioned problems.

This observation explains why residual bounds and solution-error bounds should be kept conceptually distinct. A method may preserve the least-squares objective accurately while still producing a noticeably perturbed minimizer if the problem is geometrically sensitive. That distinction becomes central later when sketching and floating-point perturbations are analyzed together.

1.4 Floating-Point Recap for Least Squares

The mixed-precision chapter develops the full floating-point model later, but the least-squares background already needs one simple principle. Forming the reduced operator, the normal matrix, or the orthogonal factorization in finite precision does not solve the exact problem posed by A and b; it solves a nearby problem. The size of this nearby perturbation depends on the working precision, the problem dimensions, and the stability of the chosen factorization. That is why the choice between normal equations, QR, SVD, and later randomized surrogates is not only a question of flop counts but also of how perturbations are amplified through the conditioning of the underlying operator [2].

1.5 Symmetric Positive Definite Operators

The covariance-transport part of the thesis is built from symmetric positive semidefinite and positive definite matrices. A matrix $A \in \mathbb{R}^{p\times p}$ is symmetric positive semidefinite if

$$x^\top Ax \geq 0 \qquad \forall x \in \mathbb{R}^p,$$

and symmetric positive definite if the inequality is strict for every nonzero x. In the latter case, A is invertible and all of its eigenvalues are strictly positive. Since the matrices are symmetric, they admit an orthogonal spectral decomposition

$$A = Q\Lambda Q^\top,$$

where Q is orthogonal and Λ is diagonal with real entries.

Empirical covariance matrices provide the main examples later on. If $Z \in \mathbb{R}^{n\times p}$ is a centered data matrix, then

$$C = \frac{1}{n} Z^\top Z$$

is symmetric positive semidefinite. In high-dimensional settings it may be singular or nearly singular, which is why regularization by λI is not an optional detail but a structural part of the method.

1.6 Matrix Square Roots and Regularization

If A is symmetric positive definite with spectral decomposition $A = Q\Lambda Q^\top$, its matrix square root and inverse square root are defined by functional calculus as

$$A^{1/2} = Q\Lambda^{1/2}Q^\top, \qquad A^{-1/2} = Q\Lambda^{-1/2}Q^\top.$$

These matrices are again symmetric positive definite, and they are the unique positive definite solutions of

$$A^{1/2}A^{1/2} = A, \qquad A^{-1/2}AA^{-1/2} = I.$$

Regularization shifts the spectrum away from zero. If C is positive semidefinite and $\lambda > 0$, then

$$C + \lambda I$$

is positive definite, with eigenvalues $\mu_i(C) + \lambda$. This shift provides a numerical safety margin for the inverse square root and controls how strongly small covariance eigenvalues are amplified.

1.7 Basic Perturbation Facts

The later covariance-transport analysis repeatedly uses simple spectral perturbation facts. The first is positivity under perturbation: if A is symmetric positive definite and E is symmetric, then

$$\lambda_{\min}(A + E) \geq \lambda_{\min}(A) - \|E\|_2.$$

Consequently, if $\|E\|_2 < \lambda_{\min}(A)$, the perturbed matrix remains positive definite. This is the basic condition that allows the square root and inverse square root of a computed approximation to remain well defined.

The second fact is qualitative rather than exact: matrix square roots and inverse square roots inherit the conditioning of the underlying spectrum. Perturbations of $A^{1/2}$ are controlled by the distance of the spectrum from zero, while perturbations of $A^{-1/2}$ are more sensitive because the scalar map $t \mapsto t^{-1/2}$ has derivative $-\frac{1}{2} t^{-3/2}$. This is the analytic source of the regularization tradeoff in covariance transport. Increasing λ improves stability of the inverse square root, but it also changes the operator that is being transported.

1.8 Subspace Embeddings and Dimensionality Reduction

The idea of replacing a large least-squares problem by a smaller one rests on a geometric requirement: the sketching matrix should approximately preserve norms and angles on the relevant subspace. This requirement is captured by the notion of an oblivious subspace embedding (OSE).

Definition 1.1 (Oblivious Subspace Embedding)**.** Let $\epsilon \in (0, 1)$ and $\delta \in (0, 1)$. A distribution $\mathcal{S}$ over $\mathbb{R}^{s \times n}$ is a (d, ϵ, δ)-OSE if, for every fixed subspace $\mathcal{V} \subseteq \mathbb{R}^n$ of dimension at most d,

$$\Pr_{S \sim \mathcal{S}}\Big[\forall\, v \in \mathcal{V} :\ (1 - \epsilon)\|v\|_2^2 \leq \|Sv\|_2^2 \leq (1 + \epsilon)\|v\|_2^2\Big] \geq 1 - \delta.$$

The squared-norm convention is natural because it is equivalent to controlling the spectral norm of the Gram distortion $U^\top S^\top SU - I$ on an orthonormal basis U of the embedded subspace. It immediately implies $\sqrt{1-\epsilon}\,\|v\|_2 \leq \|Sv\|_2 \leq \sqrt{1+\epsilon}\,\|v\|_2$.

The OSE property is a uniform Johnson–Lindenstrauss statement on subspaces. The Johnson–Lindenstrauss lemma itself provides the foundational concentration result for finite point sets:

Lemma 1.2 (Johnson–Lindenstrauss). Let $\epsilon \in (0,1)$ and let $\mathcal{P}$ be a finite set of p points in $\mathbb{R}^n$. If $S \in \mathbb{R}^{s\times n}$ has entries drawn independently from $\mathcal{N}(0,1/s)$ and

$$s \geq \frac{C \log p}{\epsilon^2},$$

then with probability at least $1-1/p$,

$$(1-\epsilon)\|u-v\|_2^2 \leq \|S(u-v)\|_2^2 \leq (1+\epsilon)\|u-v\|_2^2$$

for every pair $u, v \in \mathcal{P}$.

The passage from pointwise concentration (JL) to uniform subspace control (OSE) proceeds by a net argument: a γ-net of the unit ball in a d-dimensional subspace has cardinality at most $(3/\gamma)^d$; applying JL to this net and using the Lipschitz continuity of the sketching operator yields the subspace embedding property.

These definitions are used extensively in Chapter 3→p.13, where specific sketch families (Gaussian, Rademacher, SRHT) are shown to satisfy the OSE condition with concrete sketch-size bounds. The present section merely fixes the geometric language; the algorithmic and probabilistic details belong to the specialized chapters.

1.9 Why This Background Matters

These spectral facts extend the least-squares viewpoint from compressed rectangular operators to self-adjoint covariance operators. In the least-squares chapters, the decisive question is whether sketching preserves the geometry of a range space well enough that the reduced problem remains faithful to the original one. In the covariance-transport chapter, the decisive question becomes whether a randomized and finite-precision approximation of the covariance operator remains accurate enough after it is processed by matrix square roots and inverse square roots. The mathematical language is therefore the same in spirit, but the operator calculus is richer.

1.10 Least-Squares Perturbation Bounds

For the least-squares chapters, a more detailed perturbation statement is needed than the qualitative bound given earlier. The classical results of Wedin and Stewart provide the right framework.

Theorem 1.3 (Wedin). Let $A, \tilde{A} \in \mathbb{R}^{n\times d}$ with $n \geq d$, and suppose A has full column rank. Let $x_\star$ and $\tilde{x}_\star$ be the least-squares solutions of

$$\min_x \|Ax-b\|_2, \qquad \min_x \|\tilde{A}x-\tilde{b}\|_2.$$

Write $r_\star = b - Ax_\star$ and $\tilde{r}_\star = \tilde{b} - \tilde{A}\tilde{x}_\star$ for the optimal residuals. Then

$$\frac{\|\tilde{x}_\star - x_\star\|_2}{\|x_\star\|_2} \le \kappa_2(A)\left(\frac{\|\Delta A\|_2}{\|A\|_2} + \frac{\|\Delta b\|_2}{\|b\|_2}\right) + \kappa_2(A)^2\, \frac{\|r_\star\|_2}{\|A\|_2\,\|x_\star\|_2}\, \frac{\|\Delta A\|_2}{\|A\|_2} + O\big(\|\Delta\|^2\big),$$

where $\Delta A = \tilde{A} - A$, $\Delta b = \tilde{b} - b$, and $\|\Delta\|$ denotes the maximum relative perturbation magnitude.

The qualitative content is the same as the earlier schematic bound, but Theorem 1.3 →p.6 makes the constant structure explicit. The first term reflects the direct sensitivity of the solution to data perturbations, amplified by $\kappa_2(A)$. The second term reflects the additional sensitivity that arises when the problem is inconsistent: the residual $r_\star$ is not zero, and the perturbation of the residual direction is amplified by $\kappa_2(A)^2$.

This distinction between the consistent and inconsistent cases is central to the thesis. In sketch-and-solve methods, the sketch replaces A by a compressed operator SA. The perturbation $\Delta A = SA - A$ (in the appropriate metric) is governed by the embedding distortion ϵ, but its effect on the solution is filtered through $\kappa_2(A)$ and the residual size. Therefore, a small embedding distortion does not guarantee a small solution error unless the original problem is well-conditioned or nearly consistent. The experiments track both residual ratio and relative solution error precisely because these two quantities probe different aspects of the perturbation structure.

In the covariance-transport chapters, the same pattern reappears in a different algebraic form. The perturbation of the transport map $T_\lambda = A^{-1/2}B^{1/2}$ is not governed by $\kappa_2(A)$ alone, but by the conditioning of the regularized matrix functions $A^{-1/2}$ and $B^{1/2}$. The inverse-square-root map has a derivative proportional to $\lambda_{\min}^{-3/2}$, so the effective amplification factor is

$$K_\lambda(C_X, C_Y) \sim \frac{1}{\lambda_{\min}(C_X + \lambda I)^{3/2}}\,,$$

which is the direct analogue of $\kappa_2(A)^2$ in the least-squares setting. Regularization λ plays the same structural role as improving the conditioning of the least-squares problem: it controls the amplification factor at the cost of changing the operator being approximated.

2 Mixed-Precision Arithmetic

Mixed-precision arithmetic means that one algorithm uses more than one floating-point format. The idea is simple. One tries to perform the expensive part of the computation in a cheaper arithmetic and keeps the more accurate format for the numerically sensitive stages. In numerical linear algebra, however, this is not only a matter of implementation. A change of precision changes the perturbation model of the algorithm itself, and therefore changes the map that is actually computed [3] [2].

This chapter introduces the basic concepts needed later for randomized least-squares solvers. In particular, we discuss floating-point representation, rounding, conditioning, and the way mixed precision interacts with algorithmic structure. The main question for the later chapters is the following: if the algorithm already contains approximation through randomized sketching, how much additional perturbation from lower precision can be tolerated?

2.1 Floating-Point Model

Floating-point arithmetic replaces the continuum of real numbers by a finite set of representable values. A normalized floating-point number has the schematic form

$$x = \pm m\beta^e,$$

where β is the base, m is the significand, and e is the exponent. Since most real numbers are not exactly representable, every arithmetic operation involves rounding. Thus finite precision is not a small correction to an exact algorithm. It is the environment in which the algorithm exists.

For rounding to nearest, the standard model writes one basic operation as

$$\mathrm{fl}(x \circ y) = (x \circ y)(1 + \delta), \qquad |\delta| \leq u,$$

where u is the unit roundoff of the working format. Lower precision means a larger u, a smaller dynamic range, and therefore a smaller safety margin against cancellation, overflow, and accumulation of rounding errors. The important point is that changing precision changes the size of the perturbation allowed at every arithmetic step. This is why low precision can be attractive computationally and dangerous numerically at the same time [2] [4] [5].

The standard rounding model is useful because it allows us to interpret the computed algorithm as a nearby problem. Once we pass from one scalar operation to inner products, matrix-vector products, or matrix-matrix products, the perturbations are no longer controlled only by u, but also by the dimensions and the algebraic structure of the computation. For least-squares problems, this is especially important. Residuals, orthogonal factorizations, normal equations, and sketching operations all inherit their own precision-dependent perturbations.

2.2 Why Mixed Precision Matters

Mixed precision has become important because current hardware does not treat all precisions equally. Modern CPUs and especially accelerators often provide much higher throughput for half- or single-precision kernels than for double precision, while also reducing memory traffic and storage costs. Therefore, what used to be an implementation detail has become a design question for algorithms themselves [3] [2].

The real appeal, however, is algorithmic. Many linear algebra procedures are not equally sensitive to rounding. Bulk matrix products may tolerate lower precision, whereas factorizations, orthogonalization, residual computations, or correction solves may require a more accurate format. Mixed precision is therefore best understood as a way of deciding where accuracy is needed and where it is not. One does not choose only how much arithmetic is performed, but also where the larger perturbations are allowed to enter the computation.

2.3 Error Sources and Conditioning

In linear algebra computations, several error sources act at the same time. The data may already be perturbed when stored in low precision. Each arithmetic operation contributes a local rounding error. These perturbations accumulate through matrix products, factorizations, and iterative processes. Finally, the whole effect is filtered through conditioning. A well-conditioned problem may absorb the perturbation harmlessly, while an ill-conditioned one may amplify it strongly. Therefore, forward error, backward error, and conditioning must be kept distinct. Mixed precision changes the perturbation introduced by the algorithm, while the problem itself determines how much this perturbation is amplified.

This is especially important for least-squares problems. One must distinguish between the conditioning of the approximation problem itself and the conditioning of the algebra actually used to solve it. The latter may involve normal equations, QR factorizations, augmented systems, or preconditioned iterative refinement. Recent work on mixed-precision least squares shows that lower precision can be effective, but only in conditioning regimes that still allow stable refinement or correction [6] [7] [8] [9].

2.4 Mixed Precision in Linear Algebra Algorithms

The classical example is iterative refinement. One performs the expensive solve in lower precision, computes a more accurate residual in higher precision, and uses this residual to correct the solution. This idea is old, but modern hardware has made it newly important. Foundational work showed that one can often recover double-precision accuracy while exploiting cheaper lower-precision factorization steps, and more recent GPU implementations show substantial speed gains when tensor-core arithmetic is used carefully [10] [11] [2].

For least-squares problems the situation is more subtle than for square linear systems, because orthogonality, residual quality, and conditioning of augmented formulations all matter. This is why mixed-precision least-squares solvers often combine low-precision bulk work with higher-precision residuals, QR-based corrections, or Krylov refinement steps [6] [7]. The main issue is not only whether the problem can be solved cheaply, but whether the approximation to the projection onto the range of the data operator remains sufficiently accurate.

Recent work extends this picture beyond standard least squares. Mixed-precision FGMRES-based iterative refinement has been developed for weighted least squares, and related augmented-system approaches have been proposed for constrained and generalized least-squares problems [12] [8]. The common lesson is that lower precision is most useful in the bulk linear algebra that constructs an approximate solve or preconditioner, while the residual and correction stages must remain sufficiently accurate to control the final error.

2.5 Connection to Randomized Numerical Linear Algebra

Randomized algorithms add a second approximation before floating-point effects are even considered. A sketching matrix S replaces the original least-squares problem by a reduced one involving SA and Sb. Mixed precision can then enter at several places: in storing the data, in forming the compressed operator, in solving the reduced problem, and in any refinement step that follows.

These stages need not have the same numerical requirements. If the dominant cost lies in forming SA, low-precision matrix multiplication may be attractive. If the reduced problem is ill-conditioned, however, that gain may disappear unless the subsequent solve or residual computation is more accurate. Recent work makes this point explicit. Mixed-precision analysis of sketching-based preconditioners shows that one can choose separate precisions for the sketching and factorization stages and still obtain an effective preconditioner for GMRES-based iterative refinement, provided the perturbations remain compatible with the conditioning of the least-squares problem [13]. On the implementation side, hardware-oriented work on random projection exploits the fact that the random matrix itself may tolerate reduced precision better than the data matrix or the final solve [14].

The guiding view of this thesis is therefore the following. Sketching error and floating-point error should be analyzed together as coupled perturbations of the original least-squares problem. In the sketch-and-solve setting, one asks whether the computed reduced problem still represents the original one faithfully enough. In the sketch-and-precondition setting, one asks whether lower precision preserves enough of the conditioning improvement for the downstream solver to remain effective.

2.6 Balancing Sources of Inexactness

Mixed precision is most natural when rounding is not the only source of error in the algorithm. Randomized approximation, low-rank truncation, regularization, and even finite-sample modelling choices already introduce tolerances below which additional arithmetic accuracy may not be useful. In such settings, the purpose of precision selection is not to minimize rounding error at all costs. It is to choose an arithmetic regime in which rounding remains subordinate to the inexactness already accepted elsewhere in the method. This viewpoint aligns with recent work on balancing multiple sources of inexactness rather than optimizing one numerical layer in isolation [15] [2].

This idea can be summarized schematically as

$$\eta_{\mathrm{fp}} \lesssim \eta_{\mathrm{alg}},$$

where η_{fp} denotes floating-point perturbation and η_{alg} the algorithmic error already built into the method. In a randomized setting, the more specific target is often

$$\eta_{\mathrm{fp}} \lesssim \eta_{\mathrm{rand}},$$

meaning that reduced precision is acceptable only when it does not dominate the randomized approximation error. In an application-driven setting one may relax this further to

$$\eta_{\mathrm{fp}} \lesssim \tau_{\mathrm{stat}},$$

where τ_{stat} represents statistical or modelling tolerance.

This balancing viewpoint is especially important for the later covariance-transport analysis. There, the approximation error from randomized low-rank compression is already present before the matrix square roots and inverse square roots are evaluated. Computing every preceding product in the highest available precision may therefore solve one part of the numerical problem much more accurately than the later regularized transport map or the downstream application actually warrants. The aim is not maximal local accuracy, but a numerically coherent distribution of inexactness across the pipeline.

2.7 Stochastic Rounding as an Optional Model

The standard rounding model assumes deterministic rounding to nearest. An alternative is stochastic rounding, where each rounding step introduces a zero-mean random perturbation rather than a worst-case biased one. In favorable settings, this can reduce the growth of accumulated rounding errors from $O(nu)$

to closer to $O(\sqrt{n}\,u)$ in high probability, because the random rounding errors partially cancel instead of accumulating coherently [2].

Stochastic rounding is attractive in the present setting because the randomized covariance sketch already contains algorithmic randomness. If rounding errors can be modeled as approximately unbiased perturbations, their accumulation may be less pessimistic than in worst-case deterministic rounding analysis. This is especially relevant for the dominant products

$$Z\Omega, \qquad Z^\top(Z\Omega),$$

which contain many accumulated low-precision operations.

However, stochastic rounding introduces its own complications. Reproducibility is lost unless the random rounding stream is seeded. Hardware support remains limited, and the variance of individual rounding steps can produce occasional outliers. In addition, the unbiased-accumulation argument requires assumptions about independence that may not hold for structured or correlated data.

For the present thesis, stochastic rounding is therefore treated as an optional probabilistic model, not as a baseline assumption. The main analysis uses deterministic rounding to nearest, which is the model available on all current hardware. Stochastic rounding may be revisited as an extension once the deterministic analysis is complete and the relevant conditioning regimes are understood.

2.8 Role in This Thesis

Within this thesis, mixed precision is not a side topic. It is one of the main ways in which both randomized least-squares methods and randomized covariance transport are evaluated. The central question is not whether lower precision is always safe, but where it is safe, where it is useful, and where it changes the numerical character of the method too much to justify the computational gain.

The later chapters return to this question in numerical experiments and, later, in the climate-data workflow. The aim is modest but important: to understand mixed precision as part of algorithm design, not as a final implementation tweak applied after the mathematics has already been fixed.

3 Randomized Least Squares

Randomized numerical linear algebra uses randomness as a computational resource for restructuring expensive linear algebra tasks into smaller or more informative surrogate problems. The area now covers low-rank approximation, matrix decompositions, regression, and randomized preconditioning, but overdetermined least-squares problems remain one of its clearest and most influential successes [16] [17] [18].

This chapter focuses on sketching-based least-squares methods, which provide the first model problem for the thesis. The starting point is the overdetermined problem

$$\min_{x \in \mathbb{R}^d} \|Ax - b\|_2,$$

where $A \in \mathbb{R}^{n \times d}$ has many more rows than columns. Interpreted geometrically, one seeks the point in the range of the linear map represented by A that is closest to b. When n is large, the dominant cost lies not in the abstract existence theory, but in the repeated dense linear algebra required to solve or approximate the system.

3.1 Sketch-and-Solve Principle

The basic randomized idea is to replace the original problem by a smaller one. Given a sketching matrix $S \in \mathbb{R}^{s \times n}$ with $s \ll n$, one solves

$$\min_{x \in \mathbb{R}^d} \|SAx - Sb\|_2.$$

If S approximately preserves the geometry of the subspace carrying the least-squares information, then the reduced problem can deliver a solution whose residual is close to optimal at a lower computational cost. In operator language, the sketch should act almost isometrically on the column space of A, or more cautiously on the augmented subspace associated with both A and b. This subspace-embedding viewpoint is one of the organizing principles of modern RandNLA [18] [17].

The importance of this formulation is that it separates two questions that are easy to conflate. The first is approximation: does the sketch preserve enough geometry for the reduced problem to represent the original one faithfully? The second is computation: can the sketch be formed and used quickly enough to justify replacing the classical solver? Early least-squares algorithms already showed that this tradeoff can be favorable in practice [19] [20].

3.2 Subspace Embeddings

The geometric engine behind sketching is the notion of an oblivious subspace embedding (OSE). The definition is simple but its consequences run deep.

Definition 3.1 (Oblivious Subspace Embedding). Let $\epsilon \in (0,1)$ and $\delta \in (0,1)$. A distribution $\mathcal{S}$ over $\mathbb{R}^{s\times n}$ is a (d,ϵ,δ)-oblivious subspace embedding if, for every fixed subspace $\mathcal{V} \subseteq \mathbb{R}^n$ of dimension at most d,

$$\Pr_{S\sim\mathcal{S}}\Big[\forall\, v \in \mathcal{V}:\ (1-\epsilon)\|v\|_2^2 \le \|Sv\|_2^2 \le (1+\epsilon)\|v\|_2^2\Big] \ge 1-\delta.$$

The key point is that the sketching matrix is drawn independently of the subspace. In the least-squares context, the relevant subspace is the column space of A (or the augmented space spanned by the columns of $[A\ b]$), and the OSE property guarantees that the geometry of that subspace is approximately preserved after compression.

The squared-norm convention is the cleanest one for later spectral statements, because it is equivalent to controlling the Gram operator $U^\top S^\top SU$ on an orthonormal basis U of the embedded subspace. It immediately implies the unsquared estimate

$$\sqrt{1-\epsilon}\,\|v\|_2 \le \|Sv\|_2 \le \sqrt{1+\epsilon}\,\|v\|_2,$$

so no geometric content is lost by working with squares.

The OSE condition can be interpreted as a uniform Johnson–Lindenstrauss property on subspaces. This connection is made precise by the Johnson–Lindenstrauss lemma, which provides the foundational concentration result.

Lemma 3.2 (Johnson–Lindenstrauss). Let $\epsilon \in (0,1)$ and let $\mathcal{P}$ be a finite set of p points in $\mathbb{R}^n$. If $S \in \mathbb{R}^{s\times n}$ has entries drawn independently from $\mathcal{N}(0,1/s)$ and

$$s \ge \frac{4\log p}{\epsilon^2/2 - \epsilon^3/3},$$

then with probability at least $1-1/p$,

$$(1-\epsilon)\|u-v\|_2^2 \le \|S(u-v)\|_2^2 \le (1+\epsilon)\|u-v\|_2^2$$

for every pair $u, v \in \mathcal{P}$.

The JL lemma is a statement about finite point sets, but the extension to continuous subspaces follows by a net argument. A γ-net of the unit ball in a d-dimensional subspace has cardinality at most $(3/\gamma)^d$; applying JL to this net and using the Lipschitz continuity of the sketching operator yields the subspace embedding property. The net argument is the standard bridge from pointwise concentration to uniform geometric control.

3.3 Gaussian Sketches

The first sketch family considered in this thesis is the Gaussian sketch. Its entries are independent normal random variables with variance $1/s$, so that

$$S_{ij} \sim \mathcal{N}(0, 1/s).$$

This scaling is chosen so that the sketch is isotropic in expectation. In particular, for a fixed vector $x \in \mathbb{R}^n$, one has the identity

$$\mathbb{E}\|Sx\|_2^2 = \|x\|_2^2,$$

which explains why Gaussian sketches are a natural baseline in both theory and computation.

The main theoretical advantage of the Gaussian model is that the concentration of quadratic forms under Gaussian matrices is well understood. The following result is the central embedding guarantee.

Proposition 3.3 (Gaussian OSE). Let $S \in \mathbb{R}^{s\times n}$ be a Gaussian sketch with entries $S_{ij} \sim \mathcal{N}(0, 1/s)$. For any fixed matrix $U \in \mathbb{R}^{n\times d}$ with orthonormal columns and any $\epsilon \in (0, 1)$,

$$\Pr\Big[\|U^\top S^\top SU - I_d\|_2 > \epsilon\Big] \le 2d \cdot \exp\Big(-\frac{s(\epsilon^2 - \epsilon^3)}{4}\Big).$$

In particular, if $s \ge 4(d \log d + \log(2/\delta))/\epsilon^2$, then S is a (d, ϵ, δ)-OSE.

Proof (Proof sketch). For a fixed unit vector $u \in \mathbb{R}^n$, the random variable $\|Su\|_2^2$ is a sum of s independent χ_1^2/s variables. By standard concentration inequalities for chi-squared random variables (the Laurent–Massart bound),

$$\Pr\Big[\Big|\|Su\|_2^2 - 1\Big| > \epsilon\Big] \le 2 \exp\Big(-\frac{s(\epsilon^2 - \epsilon^3)}{4}\Big).$$

To pass from a single direction to the entire subspace, cover the unit ball of $\mathrm{col}(U)$ by a γ-net of cardinality at most $(3/\gamma)^d$ with $\gamma = \epsilon/2$. A union bound over the net, combined with the Lipschitz property of the operator norm, yields the stated inequality. The dimension requirement follows by setting the failure probability to δ and solving for s. □

The logarithmic dependence on the ambient dimension n is absent: the embedding quality depends only on the intrinsic dimension d and the sketch size s. This dimension-free character is the theoretical reason why Gaussian sketches are powerful regardless of how large the original problem is.

Closely related are dense sign sketches, often called Rademacher sketches, whose entries are independent $\pm 1/\sqrt{s}$. They retain much of the same isotropic behavior while replacing Gaussian sampling by a simpler discrete distribution. The OSE guarantee for Rademacher sketches is essentially identical to Proposition 3.3 →p.15, with the same sketch-size requirement up to constants; the proof uses the Hanson–Wright inequality instead of chi-squared concentration.

3.4 The Subsampled Randomized Hadamard Transform

The third family relevant here is the subsampled randomized Hadamard transform (SRHT). Structured sketches of this type combine random signs, a fast orthogonal transform, and subsampling. They are attractive because they preserve much of the embedding behavior of dense sketches while allowing substantially faster sketch formation when the data layout permits fast transforms.

The SRHT is defined as follows. Let $n = 2^p$ for some integer p. Denote by $H_n \in \mathbb{R}^{n\times n}$ the normalized Walsh–Hadamard matrix (satisfying $H_n H_n^\top = I_n$), and let $D \in \mathbb{R}^{n\times n}$ be a diagonal matrix whose diagonal entries are independent uniform random signs. Let $R \in \mathbb{R}^{s\times n}$ be a matrix that selects s rows uniformly at random without replacement. The SRHT sketch is then

$$S = \sqrt{\frac{n}{s}}\, R\, H_n\, D.$$

The Hadamard matrix can be applied in $O(n \log n)$ operations via the fast Walsh–Hadamard transform, which makes the total cost of forming SA equal to $O(nd \log n)$ instead of the $O(nds)$ cost of a dense Gaussian sketch. The tradeoff is that the SRHT introduces additional combinatorial structure that makes the analysis more delicate.

With this normalization the SRHT is isotropic in expectation after random row selection, so that $\mathbb{E}\|Sx\|_2^2 = \|x\|_2^2$ for fixed x. This is the structured analogue of the Gaussian scaling discussed earlier.

Proposition 3.4 (SRHT Embedding). Let $S \in \mathbb{R}^{s\times n}$ be an SRHT sketch. For any fixed matrix $U \in \mathbb{R}^{n\times d}$ with orthonormal columns and any $\epsilon \in (0, 1)$,

$$\Pr\Big[\|U^\top S^\top S U - I_d\|_2 > \epsilon\Big] \le \delta$$

provided $s \ge C\, \frac{d \log n}{\epsilon^2} \log\!\Big(\frac{d \log n}{\epsilon^2 \delta}\Big)$ for a universal constant C.

The dependence on $\log n$ is the price of the structure: the SRHT needs a slightly larger sketch than the Gaussian model to achieve the same embedding quality, but the gain in sketch-formation time can be substantial. For this thesis, the contrast between Gaussian and SRHT sketches is essential. Gaussian sketches are the cleanest theoretical baseline, and SRHT-type sketches represent the first genuinely structured option whose computational appeal may become decisive for large problems. The SRHT also plays a methodological role: because it is deterministic once the random signs and row indices are fixed, it is easier to reason about in a mixed-precision setting where the randomization and the arithmetic perturbations should be kept conceptually separate.

3.5 Residual and Solution Guarantees

The subspace-embedding property leads directly to approximation guarantees for the sketched least-squares problem. The following result is the basic accuracy statement for the sketch-and-solve paradigm.

Proposition 3.5 (Sketched LS Residual Bound). Let $A \in \mathbb{R}^{n\times d}$ have full column rank, let $x_\star = \arg\min \|Ax - b\|_2$, and let $\hat{x} = \arg\min \|SAx - Sb\|_2$. If S is a $(d+1, \epsilon, \delta)$-OSE with $\epsilon < 1$, then with probability at least $1 - \delta$,

$$\|A\hat{x} - b\|_2^2 \le \frac{1+\epsilon}{1-\epsilon}\, \|Ax_\star - b\|_2^2,$$

and hence

$$\|A\hat{x} - b\|_2 \le \sqrt{\frac{1+\epsilon}{1-\epsilon}}\, \|Ax_\star - b\|_2.$$

Proof (Proof sketch). Decompose $b = Ax_\star + r_\star$ where $r_\star$ is the optimal residual. Write $\hat{x} = x_\star + h$. The embedding property on the subspace $\mathrm{col}([A\ r_\star])$ gives

$$(1-\epsilon)\|Ah + r_\star\|_2^2 \le \|S(Ah + r_\star)\|_2^2 \le (1+\epsilon)\|Ah + r_\star\|_2^2.$$

Since $\hat{x}$ minimizes $\|S(Ax - b)\|_2$, the optimality of $\hat{x}$ for the sketched problem implies

$$\|SA\hat{x} - Sb\|_2^2 \le \|SAx_\star - Sb\|_2^2 \le (1+\epsilon)\|r_\star\|_2^2.$$

Combining the two inequalities and solving for $\|A\hat{x} - b\|_2$ yields the stated bound. □

The residual guarantee is natural but not the only quantity of interest. In many applications, one also cares about the solution error $\|\hat{x} - x_\star\|_2$. A bound on the solution error requires additional control over the geometry of the problem, specifically the extent to which the residual contributes to the solution perturbation.

Remark 3.6 (Solution error versus residual error). Residual guarantees do not automatically imply a comparably small perturbation of the minimizer. In the inconsistent case, the sensitivity of $x_\star$ is additionally filtered through the conditioning of A and by the size of the optimal residual relative to the solution itself. This is why the numerical experiments track both residual ratio and relative solution error. The residual bound is the robust first statement, while a formal solution-error theorem requires more careful assumptions than are imposed here. ⌟

These observations together explain the experimental expectations: residual quality stabilizes quickly as s grows, while solution error depends on both ϵ and $\kappa_2(A)$. The transition from residual-dominated to solution-error-dominated behavior is one of the main features that the numerical experiments are designed to capture.

3.6 Conditioning of the Sketched System

Even when the sketch provides a good embedding in exact arithmetic, the numerical behavior of the reduced problem depends critically on the conditioning of SA. The following result connects the embedding quality to the spectral properties of the compressed operator.

Proposition 3.7 (Conditioning of the Sketched Matrix)**.** Let $A \in \mathbb{R}^{n \times d}$ have full column rank with singular values $\sigma_1 \geq \cdots \geq \sigma_d > 0$, and let S be a (d, ϵ, δ)-OSE. Then with probability at least $1 - \delta$,

$$\sqrt{1-\epsilon}\,\sigma_d \leq \sigma_{\min}(SA) \leq \sigma_{\max}(SA) \leq \sqrt{1+\epsilon}\,\sigma_1,$$

and therefore

$$\kappa_2(SA) \leq \sqrt{\frac{1+\epsilon}{1-\epsilon}}\,\kappa_2(A).$$

This is a direct consequence of the embedding property applied to the left and right singular vectors of A. The important qualitative message is that a good subspace embedding cannot dramatically worsen the condition number: if ϵ is small, the sketched system is only slightly less well-conditioned than the original.

However, the proposition assumes exact arithmetic. In finite precision, the situation is more nuanced. The computed sketch $\mathrm{fl}(SA)$ is a perturbation of the ideal SA, and the effective condition number of the computed system may be larger than $\kappa_2(SA)$ by an amount that depends on the precision of the sketch-forming arithmetic and on the dimensions n and s. This is precisely where the interaction between randomized approximation and floating-point perturbation becomes important, and it motivates the mixed-precision analysis in Chapter 2→p.8.

In practice, the condition number of SA is a useful diagnostic. If $\kappa_2(SA)$ remains bounded as s varies, the sketch is preserving the spectral structure of the original problem; if it grows sharply, the sketch dimension may be insufficient, or the sketch may be failing to embed the relevant subspace. The numerical experiments in Chapter 5→p.33 track $\kappa_2(SA)$ as a function of s for exactly this reason.

3.7 From Embedding to Solver

The sketching matrix is only one part of the algorithm. Once the reduced operator SA has been formed, one still has to solve the compressed least-squares problem in a numerically sensible way. This is where conditioning re-enters the story. A good embedding should preserve not only residual geometry but also enough spectral structure that the reduced problem remains well behaved.

In practical randomized least-squares solvers, two related paradigms appear repeatedly. The first is sketch-and-solve, in which one directly solves the compressed problem. The second is sketch-and-precondition, in which the sketch is used to build a preconditioner for a more accurate iterative method or refinement procedure. Blendenpik is one of the landmark examples: it combines randomized mixing with a high-quality dense least-squares solve and shows that randomized preprocessing can improve performance without abandoning numerical linear algebra discipline [20]. More recent work goes further and shows that randomized least-squares solvers can be designed to be forward stable, and in some formulations even backward stable, rather than merely fast on average [21] [22].

This stability question is especially important for the present thesis. It is not enough to know that a sketch preserves the least-squares objective in exact arithmetic. One also needs to know how the conditioning of SA, and later of any preconditioned or refined system, interacts with the finite-precision arithmetic discussed in Chapter 2→p.8. In that sense, modern RandNLA has moved beyond the initial question of whether randomization can accelerate least squares; it now asks under what conditions randomized solvers can be trusted as serious numerical methods [23] [17].

3.8 Sketch-and-Precondition

The sketch-and-solve viewpoint is conceptually clean, but it is not always the best numerical endpoint. A second paradigm uses the sketch to build a preconditioner for the original problem rather than treating the reduced problem as the final approximation. In broad terms, one computes a factorization of the compressed matrix SA, uses that factorization to define a right preconditioner R^{-1}, and then solves the transformed problem

$$\min_y \|AR^{-1}y - b\|_2, \qquad x = R^{-1}y,$$

typically by an iterative method or by refinement around a high-quality residual computation.

This shift in viewpoint matters because the sketch no longer bears the entire burden of solution accuracy. Its role is to improve the conditioning of the operator seen by the downstream solver. If SA captures the geometry of the column space well enough, then the preconditioned matrix AR^{-1} can become substantially better behaved than the original least-squares system. Blendenpik is the classical example of this philosophy: randomized mixing and sampling are used to construct a preconditioner, but the final method remains firmly tied to stable least-squares solving rather than replacing it by a one-shot compressed surrogate [20].

At the same time, recent analysis shows that the numerical details of this paradigm matter more than one might expect. Meier, Nakatsukasa, Townsend, and Webb show that the commonly used form of sketch-and-precondition can lose numerical stability on ill-conditioned problems, even though the underlying sketch may be geometrically accurate in exact arithmetic [24]. Their analysis makes a crucial distinction between having a good randomized preconditioner in principle and embedding that preconditioner into a floating-point algorithm in a stable way. In particular, they advocate solving the preconditioned least-squares problem with an unpreconditioned iterative method applied to the transformed system, rather than relying on the numerically fragile form of the standard algorithm.

This observation sharpens the role of conditioning in the thesis. The object of interest is not merely whether $\kappa_2(SA)$ is moderate, but whether the entire solver architecture built around SA respects that favorable conditioning once rounding errors enter. It is precisely this distinction that motivates the later mixed-precision experiments. A lower-precision sketching stage may be acceptable if the resulting preconditioner still places the downstream method in a benign regime; it is unacceptable if the arithmetic perturbation destroys the conditioning improvement that the randomized step was supposed to provide.

For the present thesis, this paradigm is important for two reasons. First, it clarifies that randomized least squares is not exhausted by direct sketch-and-solve methods; preconditioning is equally central to the modern literature. Second, it provides a natural bridge to mixed precision. Recent work by Carson and Daužickaitė studies precisely this issue for sketching-based preconditioners combined with GMRES-based iterative refinement, showing how separate precision choices in the sketching and factorization stages affect whether the preconditioner remains effective [13]. This is exactly the kind of coupled approximation-and-arithmetic question that later experiments should address.

3.9 A First Experiment

The initial computational study in this thesis is deliberately simple. A random matrix $A \in \mathbb{R}^{n\times d}$ is generated together with a reference solution $x_\star$, and the right-hand side is taken as either

$$b = Ax_\star$$

in the consistent case or

$$b = Ax_\star + e$$

in the noisy case, where e is a perturbation vector. The current Octave prototype uses Gaussian sketches, while the R package extends the same benchmark structure to Rademacher sketches and the SRHT. For each sketch size s, several independent sketches are drawn, the sketched least-squares problems are solved, and the following quantities are recorded:

- the mean absolute error of the averaged solution,
- the residual ratio relative to the unsketched least-squares solution,
- the relative solution error with respect to a reference solve,
- the condition number of the sketched matrix SA.

This experiment plays an important organizational role. Mathematically, it gives a concrete setting in which subspace-embedding heuristics can be compared against observed residual and solution behavior. Computationally, it provides a common benchmark across the current prototype implementations, which already cover Gaussian, Rademacher, and SRHT sketches. In that sense, it is the smallest setting in this project where geometric approximation, conditioning of the compressed operator, and measured computation are meant to line up exactly.

3.10 Why This Experiment Matters

Even before formal guarantees are introduced, the experiment highlights the main questions that drive the later chapters. How large must the sketch dimension s be before the residual becomes reliably close to optimal? How does the conditioning of the compressed operator SA change as s varies? Which sketch family gives the best compromise between simplicity, structure, and numerical behavior? Once mixed precision is introduced, the same framework also makes it possible to ask which arithmetic stages tolerate lower precision and which do not.

The main topics to be developed further are therefore clear: subspace embeddings as the geometric core of sketching, Gaussian and structured sketches as competing algorithmic choices, residual and solution guarantees for sketch-and-solve methods, and the conditioning and finite-precision behavior of the reduced systems. Those topics place the present thesis within the broader RandNLA literature while keeping the attention on the concrete least-squares problems that the code and experiments in this repository already target.

At the same time, least squares is not the final mathematical endpoint of the thesis. It is the first model problem in which sketching, conditioning, and finite precision interact transparently. In the later covariance-transport chapter, the same perturbation viewpoint is transferred from sketched least-squares operators to empirical covariance operators and the matrix functions built from them. The role of the present chapter is therefore foundational: it supplies the geometric and numerical language that the dependence-correction analysis will reuse in a more application-aware setting.

4 Randomized Covariance Transport

This chapter formulates the climate-facing problem in the thesis that is closest to its numerical core. The climate-model application matters, but the main mathematical object is not the full post-processing workflow. It is a regularized dependence-correction map built from empirical covariance operators after marginal adjustment. This viewpoint keeps the discussion anchored in randomized numerical linear algebra and mixed-precision perturbation analysis while still preserving a clear connection to multivariate bias correction.

Existing multivariate bias-correction methods already show why dependence structure matters. Quantile-delta mapping (QDM) corrects marginal distributions while preserving modeled changes in quantiles [25]. Methods such as MBCp and MBCr combine marginal correction with dependence adjustment, and MBCn and later rank-based approaches push further toward correcting the full multivariate distribution [26] [27] [28] [29]. At the same time, the climate literature repeatedly warns that bias correction should not be treated as a black-box distributional repair, especially when the method modifies structures that matter dynamically or physically [30]. For the present thesis, this leads to a narrower and more analyzable question: can one replace an expensive dependence-correction step by a randomized low-rank approximation whose finite-precision behavior can still be controlled?

The main claim of this chapter is therefore methodological. Rather than analyze the full nonlinear climate workflow at once, we isolate a covariance-based transport problem in a latent dependence space. This reduction is still rich enough to capture the main tradeoffs between sketch dimension, regularization, conditioning, and arithmetic precision, but it is much cleaner than attempting a direct perturbation analysis of a fully nonlinear multivariate adjustment scheme.

4.1 Latent Dependence Representation

Let

$$X_h \in \mathbb{R}^{n\times p}, \qquad X_f \in \mathbb{R}^{m\times p}, \qquad Y_h \in \mathbb{R}^{n\times p}$$

denote historical climate-model data, future climate-model data, and historical reference data, respectively. The column dimension p may encode several physical variables, nearby locations, lags, seasonal features, or any local multivariate state used for downstream post-processing.

The first stage remains standard. Each marginal series is corrected by a univariate method of QDM type,

$$\widetilde{X}_h, \widetilde{X}_f = \mathrm{QDM}(X_h, X_f; Y_h),$$

so that the eventual output inherits the desired marginal behavior from the reference historical data together with modeled changes in quantiles [25]. What remains is the dependence correction.

To express dependence in a form closer to linear algebra, it is convenient to pass to a latent Gaussianized space. The precise standardization map can be chosen in several closely related ways, but it should avoid empirical probabilities equal to 0 or 1, since Φ^{-1} would then produce infinities. A convenient finite definition is obtained from plotting-position ranks. For each column j, write

$$u_{X,ij} = \frac{\mathrm{rank}(\widetilde{X}_h(i,j)) - \frac{1}{2}}{n}, \qquad u_{Y,ij} = \frac{\mathrm{rank}(Y_h(i,j)) - \frac{1}{2}}{n},$$

and similarly

$$u_{f,ij} = \frac{\mathrm{rank}(\widetilde{X}_f(i,j)) - \frac{1}{2}}{m}$$

for the future sample. Ties are handled by average ranks; any fixed deterministic tie-breaking rule would serve equally well. We then define

$$Z_X(i,j) = \Phi^{-1}(u_{X,ij}), \qquad Z_Y(i,j) = \Phi^{-1}(u_{Y,ij}),$$

and similarly obtain Z_f from u_f. In this representation, marginal correction is separated from dependence correction: the marginals are handled before and after the latent transform, while the multivariate adjustment acts on the standardized latent variables.

This reduction is not meant to claim that all relevant climate dependence is Gaussian. It is a modeling step that isolates a dependence operator that is computationally tractable and mathematically interpretable. In particular, it allows the dependence-correction stage to be expressed through covariance operators in a finite-dimensional inner-product space.

4.2 Regularized Covariance Transport

Assume that the columns of Z_X and Z_Y have been centered. Their empirical covariance operators are

$$C_X = \frac{1}{n} Z_X^\top Z_X, \qquad C_Y = \frac{1}{n} Z_Y^\top Z_Y.$$

The dependence-correction step is then represented by the regularized transport map

$$T_\lambda = (C_X + \lambda I)^{-1/2}(C_Y + \lambda I)^{1/2},$$

where $\lambda > 0$ is a regularization parameter. In the language of optimal transport, this is a *regularized whitening–coloring covariance transport*: the map first whitens the source covariance (inverse square root) and then colors the result with the target covariance (square root). It coincides with the Bures–Wasserstein optimal-transport map when C_X and C_Y commute, but in general it is not the symmetric optimal-transport map. The present thesis adopts it primarily for its algebraic tractability and its natural connection to matrix-function perturbation theory, not as a claim of optimality in the transport sense.

The role of λ is both numerical and conceptual. Numerically, it keeps the source and target operators strictly positive definite and prevents the inverse square root from reacting catastrophically to small singular values or nearly redundant variables. Conceptually, it acknowledges that in high-dimensional post-processing tasks one should not insist on recovering dependence structure below the scale at which the empirical covariance is already poorly determined.

The corrected latent future data are then obtained as

$$Z_f^{\mathrm{corr}} = Z_f T_\lambda.$$

Remark 4.1 (Regularization bias)**.** For $\lambda = 0$, the idealized transport map $T = C_X^{-1/2} C_Y^{1/2}$ satisfies

$$T^\top C_X T = C_Y$$

whenever the inverse square root is well defined. For $\lambda > 0$, the regularized map instead satisfies

$$T_\lambda^\top (C_X + \lambda I) T_\lambda = C_Y + \lambda I,$$

and therefore transports the regularized covariance exactly, not the original unregularized covariance. Thus λ controls both numerical stability and modelling bias. ⌟

Finally, because a linear map in latent space will generally disturb the exact marginal distributions, the latent correction is used only to update ranks. For each variable, the values from the QDM-corrected future sample $\widetilde{X}_f(:, j)$ are sorted and then rearranged according to the ranks of $Z_f^{\mathrm{corr}}(:, j)$. The final output therefore combines

$$\text{marginals from QDM} \quad + \quad \text{dependence from covariance transport.}$$

This construction should be viewed as a lower-rank and more analyzable alternative to full multivariate distribution matching. It is less ambitious than MBCn in what it tries to capture, but it is correspondingly better aligned with the perturbation tools of numerical linear algebra.

4.3 Randomized Low-Rank Approximation

The direct formation and decomposition of C_X and C_Y is unattractive when p is large. The thesis therefore proposes to approximate the covariance operators by randomized low-rank surrogates. Let $\Omega \in \mathbb{R}^{p \times \ell}$ be a sketching matrix with $\ell = r + q$, where r is the target rank and q an oversampling parameter. Then the dominant range of C_X may be sampled through

$$Y_X = C_X \Omega = \frac{1}{n} Z_X^\top (Z_X \Omega),$$

and analogously for C_Y. This is precisely the type of factorization-friendly structure for which randomized low-rank approximation is effective [16] [17].

If Q_X is an orthonormal basis for the range of Y_X, the covariance operator is approximated by

$$C_X \approx Q_X B_X Q_X^\top, \qquad B_X = Q_X^\top C_X Q_X,$$

with an analogous approximation for C_Y. The transport map is then built from

$$\widehat{C}_{X,\lambda} = Q_X B_X Q_X^\top + \lambda I, \qquad \widehat{C}_{Y,\lambda} = Q_Y B_Y Q_Y^\top + \lambda I.$$

The computational advantage is immediate: the numerically demanding work is concentrated in matrix products of the form $Z\Omega$ and $Z^\top(Z\Omega)$, while the eigenanalysis is transferred to the small dense matrices B_X and B_Y.

Proposition 4.2 (Randomized covariance approximation)**.** Let $C \in \mathbb{R}^{p\times p}$ be symmetric positive semidefinite, and let Q be produced by a randomized range finder with target rank r and oversampling parameter q. Then the compression error

$$\|(I - QQ^\top)C\|_2$$

is controlled by the spectral tail of C beyond rank r. In particular, standard range-finder bounds imply that there exist oversampling-dependent constants $c_{r,q}$ and $d_{r,q}$ such that

$$\mathbb{E}\,\|(I - QQ^\top)C\|_2 \le c_{r,q}\,\mu_{r+1}(C) + d_{r,q}\left(\sum_{j>r}\mu_j(C)^2\right)^{1/2},$$

where $\mu_1(C) \ge \mu_2(C) \ge \cdots \ge 0$ are the eigenvalues of C in nonincreasing order Chs. 10–11 [16] Sec. 8 [17].

Proof (Proof sketch)**.** This is the standard randomized range-finder estimate specialized to a symmetric positive semidefinite matrix. Since C is self-adjoint, its singular values are precisely the eigenvalues $\mu_j(C)$. □

Moreover, the regularized matrix functions of a low-rank surrogate can be applied without forming a full dense $p \times p$ matrix. If $Q \in \mathbb{R}^{p\times r}$ has orthonormal columns and $B \in \mathbb{R}^{r\times r}$ is symmetric positive semidefinite, then

$$(QBQ^\top + \lambda I)^{-1/2} = \lambda^{-1/2} I + Q\Big((B + \lambda I_r)^{-1/2} - \lambda^{-1/2} I_r\Big)Q^\top,$$

and similarly

$$(QBQ^\top + \lambda I)^{1/2} = \lambda^{1/2} I + Q\Big((B + \lambda I_r)^{1/2} - \lambda^{1/2} I_r\Big)Q^\top.$$

These identities show that the transport layer only needs matrix functions of the small core matrix $B + \lambda I_r$, together with applications of Q and $Q^\top$.

The point is not only efficiency. A low-rank approximation introduces an explicit randomized error scale. Once this scale is visible, it becomes meaningful to ask whether lower precision is acceptable because its rounding error remains smaller than the approximation error already introduced by the randomized compression.

4.4 Independent and Shared Reduced Bases

The randomized low-rank approximation described above produces an orthonormal basis Q_X for the dominant range of C_X and a basis Q_Y for the dominant range of C_Y. When the source and target covariance operators are approximated independently, the resulting surrogates take the form

$$C_X \approx Q_X B_X Q_X^\top, \qquad C_Y \approx Q_Y B_Y Q_Y^\top,$$

where $B_X = Q_X^\top C_X Q_X$ and $B_Y = Q_Y^\top C_Y Q_Y$ are small dense core matrices. This independent-basis construction is the natural output of two separate range-finder calls.

The difficulty is that Q_X and Q_Y may span different reduced coordinate systems. Even if each covariance is approximated reasonably well in isolation, the transport map combines the inverse square root of one surrogate with the square root of the other:

$$\widehat{T}_\lambda = (Q_X B_X Q_X^\top + \lambda I)^{-1/2}(Q_Y B_Y Q_Y^\top + \lambda I)^{1/2}.$$

When $Q_X \neq Q_Y$, the matrix functions are evaluated in different subspaces, and their product may be less stable than the individual factors suggest. This coordinate mismatch does not necessarily cause large errors, but it removes the algebraic simplification that would otherwise stabilize the transport map.

An alternative is the *shared-basis* variant. Instead of computing two separate range finders, one constructs a single orthonormal basis Q that approximately spans the union of the dominant ranges of C_X and C_Y. A natural construction is to form the combined sketch matrix

$$Y_{\text{joint}} = [C_X \Omega,\ C_Y \Omega]$$

and compute Q as an orthonormal basis for the range of Y_{joint}. Both covariance operators are then projected onto this shared basis:

$$C_X \approx Q B_X Q^\top, \qquad C_Y \approx Q B_Y Q^\top,$$

with $B_X = Q^\top C_X Q$ and $B_Y = Q^\top C_Y Q$. The regularized transport map becomes

$$\widehat{T}_\lambda = (Q B_X Q^\top + \lambda I)^{-1/2}(Q B_Y Q^\top + \lambda I)^{1/2},$$

and both matrix functions are now evaluated in the same reduced coordinate system. This eliminates the coordinate mismatch and is expected to reduce $\|T_\lambda - \widehat{T}_\lambda\|_2$ even when the raw covariance errors $\|C_X - \widehat{C}_X\|_2$ and $\|C_Y - \widehat{C}_Y\|_2$ are comparable to those of the independent construction.

The shared-basis variant should not be understood as universally superior. Its advantage is a hypothesis to be tested experimentally: in regimes where the low-rank approximation is already reasonably accurate, shared coordinates may stabilize the transport map enough to make precision effects visible. In regimes where the approximation itself is too coarse, neither coordinate alignment nor precision choice will rescue the method. The experiments in Chapter 5 → p.33 test this hypothesis explicitly.

4.5 Mixed Precision and Balancing Inexactness

This is the place where the climate application and the mixed-precision chapter meet most naturally. The important question is not whether lower precision is faster in the abstract. It is whether reduced precision changes the dependence correction less than the inexactness already present in the modeling pipeline. This is exactly the viewpoint advocated in recent mixed-precision work on balancing multiple sources of inexactness [15] [2].

For the proposed method, the natural arithmetic split is the following. The bulk products

$$Z\Omega, \qquad Z^\top(Z\Omega)$$

are the most plausible candidates for reduced precision. By contrast, the orthogonalization that produces Q, the small eigendecompositions of B_X and B_Y, and the evaluation of matrix square roots and inverse square roots are more sensitive and are therefore better kept in double precision. This follows the general mixed-precision pattern already visible in least-squares sketching and refinement methods: use cheaper arithmetic where the computation is large and structurally robust, and keep higher precision where instability would be amplified by conditioning [13] [6].

Accordingly, the computed covariance approximation should be written as

$$\widehat{C}_X = C_X + E_{\text{rand},X} + E_{\text{fp},X}, \qquad \widehat{C}_Y = C_Y + E_{\text{rand},Y} + E_{\text{fp},Y},$$

where E_{rand} denotes randomized approximation error and E_{fp} floating-point error. A concrete floating-point model enters already at the sample matrix level. If the dominant product is formed in lower precision with unit roundoff u_ℓ, then one may write

$$\widehat{Y}_X = \mathrm{fl}_{u_\ell}\Big(\frac{1}{n}\, Z_X^\top(Z_X\Omega)\Big) = C_X\Omega + \Delta Y_X,$$

and analogously for $\widehat{Y}_Y$. Under the standard matrix-product rounding model,

$$\|\Delta Y_X\|_2 \lesssim \gamma_n^{(u_\ell)}\, \frac{\|Z_X\|_2^2}{n}\, \|\Omega\|_2, \qquad \gamma_n^{(u_\ell)} = \frac{nu_\ell}{1 - nu_\ell},$$

with the same form for ΔY_Y [2]. Thus the floating-point contribution depends not only on the unit roundoff, but also on the accumulation length, the scale of the latent data, the norm of the sketching operator, and on whether accumulation is performed in low or high precision. The key regime is the balanced one,

$$\|E_{\text{fp}}\| \lesssim \|E_{\text{rand}}\|,$$

or more generally when the floating-point perturbation remains below the statistical or modeling tolerance of the application. In that regime, computing the sketch in full double precision would amount to solving one part of the numerical problem much more accurately than the rest of the pipeline warrants.

Proposition 4.3 (Covariance sketch product perturbation). Let $Z \in \mathbb{R}^{n\times p}$ be a centered data matrix with empirical covariance $C = \frac{1}{n} Z^\top Z$, let $\Omega \in \mathbb{R}^{p\times \ell}$ be a sketching matrix, and let the computed sample product be formed in working precision with unit roundoff u. Write

$$\widehat{Y} = \mathrm{fl}_u\Big(\frac{1}{n}\, Z^\top (Z\Omega)\Big) = C\Omega + \Delta Y.$$

Then, under the standard inner-product rounding model with accumulation in precision u,

$$\|\Delta Y\|_2 \le \frac{nu}{1-nu}\,\frac{\|Z\|_2^2}{n}\,\|\Omega\|_2,$$

provided that $nu < 1$. In particular, writing $\eta_{\mathrm{fp}} = \|\Delta Y\|_2/\|\Omega\|_2$, one obtains the operator-norm estimate

$$\eta_{\mathrm{fp}} \lesssim u\,\|Z\|_2^2 = u\,n\,\|C\|_2 = u\,n\,\lambda_{\max}(C),$$

so that the floating-point perturbation scales with the unit roundoff, the sample size entering the accumulation, and the spectral scale of the empirical covariance.

Proof. This is a direct application of the standard matrix-product rounding model Thm. 3.5 [2]. Each column of $\frac{1}{n} Z^\top (Z\Omega)$ is an inner product of length n, so the columnwise relative perturbation is bounded by $\gamma_n = nu/(1-nu)$. Taking norms and using $\|C\|_2 = \|Z\|_2^2/n$ gives the result. □

The practical content of Proposition 4.3→p.28 is that the floating-point contribution to the covariance sketch perturbation is computable from the data norms, the accumulation length, and the unit roundoff alone. The balanced regime

$$\eta_{\mathrm{fp}} \lesssim \eta_{\mathrm{rand}}$$

can therefore be tested before committing to a precision choice, by comparing η_{fp} against a randomized truncation error estimate (e.g. from a range-finder residual).

4.6 Perturbation Perspective

The central analytical task is not to study the entire bias-correction workflow at once, but to bound the perturbation of the transport map induced by the covariance approximation. Define

$$A = C_X + \lambda I, \qquad B = C_Y + \lambda I,$$

and let

$$\widehat{A} = A + \Delta A, \qquad \widehat{B} = B + \Delta B$$

be the regularized approximations produced by randomized low-rank compression and finite-precision arithmetic. Then

$$T_\lambda = A^{-1/2} B^{1/2}, \qquad \widehat{T}_\lambda = \widehat{A}^{-1/2} \widehat{B}^{1/2}.$$

The natural forward quantities are

$$\|T_\lambda - \widehat{T}_\lambda\|_2 \qquad \text{and} \qquad \|Z_f(T_\lambda - \widehat{T}_\lambda)\|_F.$$

At a first formal level one may split the transport error as

$$\|T_\lambda - \widehat{T}_\lambda\|_2 \le \|A^{-1/2}\|_2\, \|B^{1/2} - \widehat{B}^{1/2}\|_2 + \|A^{-1/2} - \widehat{A}^{-1/2}\|_2\, \|\widehat{B}^{1/2}\|_2.$$

This identity already reveals the main conditioning mechanism. Small covariance perturbations are not enough by themselves. Their effect is filtered through matrix square roots and, more severely, inverse square roots. Therefore the decisive quantity is not only the size of ΔA and ΔB, but also the spectral safety margin created by the regularization parameter λ.

If

$$\|\Delta A\|_2 \le \eta_A, \qquad \|\Delta B\|_2 \le \eta_B,$$

the later perturbation analysis can be organized into a small chain of deterministic statements.

Proposition 4.4 (Positive definiteness under perturbation)**.** Let $A = C_X + \lambda I$ and $\widehat{A} = A + \Delta A$, where A and $\widehat{A}$ are symmetric. If $\|\Delta A\|_2 < \lambda_{\min}(A)$, then $\widehat{A}$ remains symmetric positive definite and

$$\lambda_{\min}(\widehat{A}) \ge \lambda_{\min}(A) - \|\Delta A\|_2.$$

The same statement holds for B and $\widehat{B}$.

Proof (Proof sketch)**.** For any unit vector x,

$$x^\top \widehat{A} x = x^\top A x + x^\top \Delta A x \ge \lambda_{\min}(A) - \|\Delta A\|_2.$$

Taking the infimum over $\|x\|_2 = 1$ gives the bound. □

Proposition 4.5 (Perturbation of regularized square roots)**.** Assume that A, $\widehat{A}$, B, and $\widehat{B}$ are symmetric positive definite, and set

$$\alpha_0 = \min\{\lambda_{\min}(A), \lambda_{\min}(\widehat{A})\}, \qquad \beta_0 = \min\{\lambda_{\min}(B), \lambda_{\min}(\widehat{B})\}.$$

Then

$$\|A^{-1/2} - \widehat{A}^{-1/2}\|_2 \le \frac{1}{2\alpha_0^{3/2}}\, \|\Delta A\|_2,$$

and

$$\|B^{1/2} - \widehat{B}^{1/2}\|_2 \le \frac{1}{2\beta_0^{1/2}}\, \|\Delta B\|_2.$$

Proof (Proof sketch). These inequalities are standard perturbation bounds for matrix functions on the positive definite cone. One route is to write the Fréchet derivative of the square-root map through a Sylvester equation and then apply a mean-value bound for matrix functions; see Chs. 3 and 6 [31]. The inverse-square-root estimate then follows from the identity

$$A^{-1/2} - \widehat{A}^{-1/2} = A^{-1/2}(\widehat{A}^{1/2} - A^{1/2})\widehat{A}^{-1/2},$$

combined with the square-root perturbation bound. The important qualitative point is the scaling: the inverse square root is more sensitive near small eigenvalues, with amplification proportional to $\alpha_0^{-3/2}$. □

Theorem 4.6 (Perturbation of covariance transport). Assume that the hypotheses of the previous proposition hold. Then

$$\|T_\lambda - \widehat{T}_\lambda\|_2 \le \|A^{-1/2}\|_2 \, \|B^{1/2} - \widehat{B}^{1/2}\|_2 + \|A^{-1/2} - \widehat{A}^{-1/2}\|_2 \, \|\widehat{B}^{1/2}\|_2,$$

and therefore

$$\|T_\lambda - \widehat{T}_\lambda\|_2 \le \frac{\|A^{-1/2}\|_2}{2\beta_0^{1/2}} \, \eta_B + \frac{\|\widehat{B}^{1/2}\|_2}{2\alpha_0^{3/2}} \, \eta_A.$$

In particular, the amplification factor may be taken as

$$K_\lambda(C_X, C_Y, \Delta A, \Delta B) = \max\left\{ \frac{\|A^{-1/2}\|_2}{2\beta_0^{1/2}}, \frac{\|\widehat{B}^{1/2}\|_2}{2\alpha_0^{3/2}} \right\},$$

so that

$$\|T_\lambda - \widehat{T}_\lambda\|_2 \le K_\lambda \, (\eta_A + \eta_B).$$

When $\|\Delta A\|_2$ and $\|\Delta B\|_2$ are small relative to the regularized spectral gaps, this behaves like a constant depending on $\lambda_{\min}(C_X + \lambda I)$ and $\lambda_{\min}(C_Y + \lambda I)$. In particular, the inverse-square-root term deteriorates at the rate $\lambda_{\min}(C_X + \lambda I)^{-3/2}$.

Proof. Split the difference as

$$T_\lambda - \widehat{T}_\lambda = A^{-1/2}(B^{1/2} - \widehat{B}^{1/2}) + (A^{-1/2} - \widehat{A}^{-1/2})\widehat{B}^{1/2},$$

then apply the triangle inequality and the previous proposition. The final bound is first order in η_A and η_B, with constants determined by the regularized spectral gaps. □

The randomized and arithmetic contributions can then be separated as

$$\eta_A \le \eta_{\mathrm{rand},A} + \eta_{\mathrm{fp},A}, \qquad \eta_B \le \eta_{\mathrm{rand},B} + \eta_{\mathrm{fp},B}.$$

For spectra with visible decay, the randomized term is governed by the tail beyond the chosen rank, up to oversampling constants from the range-finder bound above [16] [17]. The floating-point term, by contrast, follows the sketch-product model above and is controlled by the working precision, the accumulation length, the scale of the latent data, and the norm of the sketching operator. The thesis question is precisely how these two error sources should be balanced against each other.

Algorithm 1: Mixed-precision randomized covariance transport

Input: Historical model data X_h, future model data X_f, historical reference data Y_h, target rank r, oversampling q, regularization λ, sketch precision u_ℓ

Output: Corrected future sample $\widetilde{X}_f^{\mathrm{corr}}$, diagnostics

Apply QDM-type marginal correction to obtain $\widetilde{X}_h$, $\widetilde{X}_f$;

Transform $\widetilde{X}_h$, $\widetilde{X}_f$, Y_h to latent rank-normal variables Z_X, Z_f, Z_Y;

Form randomized covariance sketches in precision u_ℓ:

$$Y_X = \tfrac{1}{n} Z_X^\top (Z_X \Omega_X), \qquad Y_Y = \tfrac{1}{n} Z_Y^\top (Z_Y \Omega_Y)$$

;

Compute orthonormal bases Q_X, Q_Y (or shared basis Q);

Form small covariance cores $B_X = Q_X^\top C_X Q_X$, $B_Y = Q_Y^\top C_Y Q_Y$;

Build regularized low-rank covariance surrogates $\widehat{C}_{X,\lambda}$, $\widehat{C}_{Y,\lambda}$;

Evaluate regularized transport map:

$$\widehat{T}_\lambda = (\widehat{C}_{X,\lambda})^{-1/2} \, (\widehat{C}_{Y,\lambda})^{1/2}$$

;

Apply $Z_f^{\mathrm{corr}} = Z_f \, \widehat{T}_\lambda$;

Restore QDM marginals by rank rearrangement;

return corrected future sample and diagnostics;

Diagnostics: covariance error $\|C - \widehat{C}\|_2$; transport-map error $\|T_\lambda - \widehat{T}_\lambda\|_2$; induced future-data perturbation $\|Z_f T_\lambda - Z_f \widehat{T}_\lambda\|_F$; ratio $\eta_{\mathrm{fp}}/\eta_{\mathrm{rand}}$; runtime;

Corollary 4.7 (Perturbation of corrected future data)**.** Under the hypotheses of the theorem,

$$\|Z_f T_\lambda - Z_f \widehat{T}_\lambda\|_F \le \|Z_f\|_F \, \|T_\lambda - \widehat{T}_\lambda\|_2.$$

Proof. This is the submultiplicativity of the Frobenius norm with the operator norm on the right. □

4.7 Algorithmic Consequences

This formulation leads to a concrete algorithmic decomposition, presented formally as 1 → p.31.

This is the natural architecture for the software contribution as well. The implementation should separate a randomized covariance-approximation layer, a transport-map layer, and a climate-facing wrapper that handles marginal correction and rank restoration. Such a decomposition keeps the numerical core reusable while preserving a clear application entry point.

The local `randMBC` prototype now realizes this decomposition explicitly. It exposes a deterministic marginal-correction and rank-restoration wrapper around a randomized covariance-transport core, together with a synthetic benchmark that compares the full regularized transport baseline, selected reference methods from the vendored `MBC` package, and the current randomized low-rank implementation. The first controlled runs are informative even though they are not yet favorable to the randomized method: on the present synthetic benchmark, the deterministic full transport and the MBC references remain markedly more accurate than the current low-rank randomized surrogate. This should not be read as a failure of the covariance-transport viewpoint itself. Rather, it identifies more precisely where the open problem lies, namely in the approximation regime where low-rank compression becomes accurate enough that transport-map perturbations no longer dominate the downstream dependence correction.

The same decomposition also determines the experimental program. The decisive sweeps are not arbitrary. They should vary the approximation rank, the regularization parameter, and the arithmetic precision while tracking covariance error, transport-map error, and the induced dependence error on the corrected future data. In particular, one should expect a regime where single precision is essentially indistinguishable from double because randomized truncation already dominates, and another regime where further rank increase reveals floating-point error because the approximation error has become too small to hide it.

4.8 Role in the Thesis

The purpose of this chapter is not to claim that covariance transport solves the whole problem of multivariate climate bias correction. It does not replace the need for application-side diagnostics, and it should not be presented as a universal substitute for richer multivariate schemes such as MBCn. Its role is narrower and more precise. It identifies a mathematically serious dependence-correction problem that is simultaneously

1. motivated by climate-model post-processing,
2. expressible in the language of randomized low-rank approximation,
3. sensitive to conditioning and regularization,
4. and naturally suited to mixed-precision analysis.

For the later parts of the thesis, this chapter therefore acts as a bridge. Upstream, it uses the perturbation viewpoint of Chapter 2 → p.8 and the randomized approximation ideas from Chapter 3 → p.13. Downstream, it provides the specific method family that the implementation and the climate-oriented experiments should test. In this sense, randomized covariance transport is the point at which the thesis becomes simultaneously application-aware and still unmistakably about numerical linear algebra.

5 Numerical Experiments

This chapter describes the experimental framework used in the repository for randomized least-squares solvers and for the later covariance-transport problem. The goal is not only to illustrate the theory, but to test whether the geometric statements from Chapter 3 →p.13 and the perturbation viewpoint from Chapter 4 →p.22 remain visible in computed residuals, computed solutions, covariance approximations, and transport errors. The experiments also provide the first controlled setting in which the floating-point questions from Chapter 2 →p.8 can be studied in the same benchmark.

The experiments use intentionally synthetic data. This is useful: by generating the data matrix, covariance structure, and reference solution directly, we separate the effect of the sketch from the complications of a downstream application. The experiments are therefore simple enough to interpret and still rich enough to show the interaction between embedding quality, conditioning, regularization, and arithmetic.

5.1 Experimental Objective

The experiments are naturally divided into two parts. Part I studies sketched least-squares solvers as the first model problem. Part II studies randomized covariance transport as the climate-facing operator problem introduced in Chapter 4 →p.22. The repository currently implements mainly the first part, but the second part already has a clear mathematical specification.

For the least-squares experiments, the main question is how the quality of a sketched solve changes as the sketch dimension varies and as the sketch family changes. The repository currently tracks four quantities:

- the mean absolute error of the averaged sketch solution,
- the residual ratio relative to a reference least-squares solve,
- the relative solution error with respect to the reference solution,
- the condition number of the sketched matrix SA.

These quantities reflect different numerical phenomena. The residual ratio measures how well the sketched problem preserves the least-squares objective. The solution error is more sensitive to conditioning and therefore detects failures that may not be visible at the residual level. The condition number of SA is not an error measure by itself, but it often explains why two sketches with similar residual behavior may differ strongly at the level of the computed solution.

The experiments are therefore not only comparisons of random projections. They test how approximation quality, solver sensitivity, and arithmetic interact in one concrete computational pipeline.

5.2 Synthetic Least-Squares Model

The current implementations use one common synthetic model. A matrix $A \in \mathbb{R}^{n\times d}$ is generated with independent standard normal entries, and the reference solution is chosen as

$$x_\star = \mathbf{1} \in \mathbb{R}^d.$$

The right-hand side is then formed as

$$b = Ax_\star + e,$$

where e is either zero or a Gaussian perturbation scaled by a prescribed noise level. In this way, both the consistent and noisy regimes are available within the same template.

For each generated instance, a reference least-squares solution x_{ref} is computed from the full system. In the R implementation this is done with `qr.solve`, while the Octave prototypes use the backslash solver. The reference residual

$$r_{\mathrm{ref}} = Ax_{\mathrm{ref}} - b$$

provides the baseline against which the sketched residuals are compared. This is important because each trial is measured against the corresponding unsketched least-squares solve.

The synthetic model is deliberately modest. It does not attempt to mimic the spectral structure of a climate-data operator or the coherence patterns of an application matrix. Its role is to isolate the basic mechanisms of randomized least squares under controlled conditions.

5.3 Implemented Sketch Families

The repository currently supports three sketch families in the main R package: Gaussian sketches, Rademacher sketches, and the SRHT. The Octave prototypes currently cover Gaussian sketches and the SRHT. This reflects the current state of the software rather than a conceptual distinction. The R package is the intended long-term implementation base, while the Octave scripts remain useful as compact prototypes.

The Gaussian sketch is the cleanest theoretical baseline. The Rademacher sketch replaces Gaussian entries by random signs and tests how much of the same behavior survives under a simpler discrete distribution. The SRHT introduces structure and therefore changes both the cost model and the embedding constants. It is the first sketch family in the project for which it is natural to ask whether the additional algebraic structure is worth the analytical complexity.

There is one practical restriction that matters experimentally. The current SRHT implementations require the row dimension to be a power of two. Therefore, the R and Octave code round n upward when needed before constructing the Hadamard-based sketch. This is a simple example of a theoretical assumption becoming an implementation condition.

5.4 Per-Trial Computation

For a fixed sketch size s, one trial proceeds as follows. A sketching matrix $S \in \mathbb{R}^{s\times n}$ is drawn, the compressed system

$$SAx \approx Sb$$

is formed, and the reduced least-squares problem is solved. The computed solution for that trial is denoted by $\hat{x}$. The experiment then records:

$$\begin{aligned} \text{residual ratio} &= \frac{\|A\hat{x} - b\|_2}{\|Ax_{\text{ref}} - b\|_2}, \\ \text{solution error} &= \frac{\|\hat{x} - x_{\text{ref}}\|_2}{\|x_{\text{ref}}\|_2}, \\ \text{condition number} &= \kappa_2(SA). \end{aligned}$$

These definitions are aligned with the theoretical discussion in Chapter 3 → p. 13. The residual ratio corresponds to the sketch-and-solve guarantee that the compressed problem should preserve the least-squares objective up to a controlled distortion. The solution error reflects the fact that preserving the residual does not automatically guarantee an equally small perturbation in the minimizer. The condition number of SA is recorded because it is one of the main mechanisms that links these two behaviors.

In addition, the implementations average the independent trials at each sketch size. If $\hat{x}^{(1)}, \ldots, \hat{x}^{(m)}$ are the trial solutions, the code forms their mean and reports the mean absolute error of this averaged estimator relative to the planted vector $x_\star$. This is not a standard RandNLA performance criterion, but it is useful here because it exposes the variability of the individual trial solutions.

5.5 Parameter Sweeps

The experiments vary the sketch dimension s over a prescribed range and repeat the randomized solve several times for each value. In the current R code, the generic routine `sketch_lstsq_experiment` accepts the parameters n, d, a vector of sketch sizes, a trial count, a noise level, and the sketch family. Specialized wrappers currently expose Gaussian and SRHT runs directly, but the underlying routine already supports all three sketch families.

The Octave scripts implement the same logic in a more direct form. The Gaussian prototype sweeps over `s_vals = 1:99` by default for a problem of size $n = 100$, $d = 10$, while the SRHT prototype uses a coarser sweep adapted to the power-of-two requirement. In both cases, several independent sketches are drawn for each s, and the recorded history stores the sketch size together with the averaged error metrics.

This repeated-trial design is essential. A single randomized sketch may be atypical, especially for small sketch sizes. Averaging over several trials does not remove randomness from the method, but it makes the observed trends much more interpretable.

5.6 Precision-Sensitivity Baseline

The mixed-precision part of the repository is still at an early stage, but the current Octave Gaussian experiment already contains a useful first comparison. It can run either in ordinary double precision or, when the symbolic package is available, in a higher-precision mode used as a surrogate for reduced rounding error. This is not a full mixed-precision implementation in the sense discussed in Chapter 2→p.8. It is a controlled way to ask whether a visible loss of accuracy is mainly geometric or mainly arithmetic.

That distinction matters. If increasing arithmetic precision leaves the curves essentially unchanged, the dominant limitation is likely the sketch itself or the conditioning of the reduced problem. If the higher-precision run materially improves the behavior, then floating-point perturbations are already visible at this synthetic scale. The current precision experiment is therefore best viewed as a diagnostic baseline for the later mixed-precision work.

5.7 Least-Squares Sketch-Size Sweep

The first implemented experiment tests how the quality of a sketched solve changes as the sketch dimension increases. The setup follows the synthetic model from Section 5.2→p.34 with $n = 1024$, $d = 20$, a planted solution $x_\star = \mathbf{1}$, and no noise. A Gaussian sketch is used, the sketch dimension s is swept over the values

$$s \in \{30, 40, 60, 80, 100, 150, 200, 300\},$$

and 20 independent trials are drawn for each value.

Figure 5.1→p.37 shows the mean residual ratio as a function of s. The expected qualitative behaviour is clearly visible: the residual ratio decreases toward one as the sketch dimension grows, with the most rapid improvement occurring for s

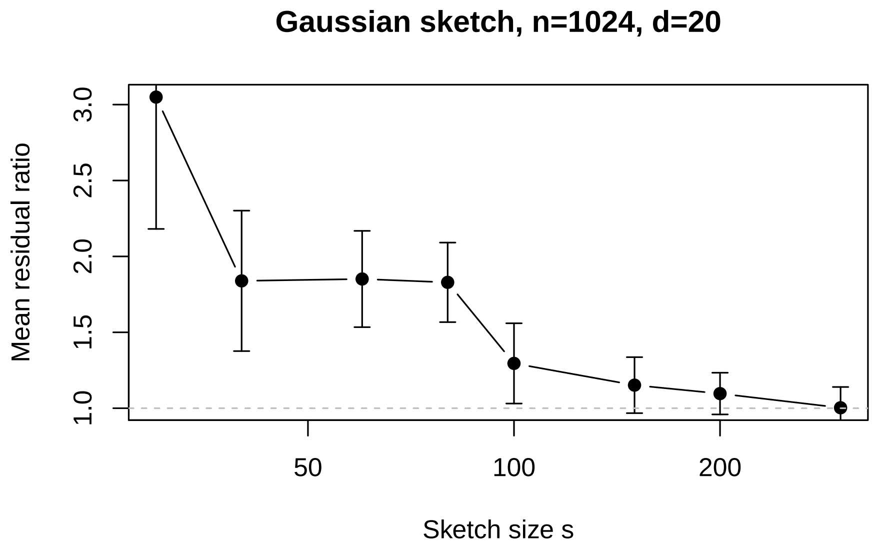

Figure 5.1: Mean residual ratio versus sketch size for Gaussian sketching of a consistent least-squares problem with $n = 1024$, $d = 20$. Error bars indicate one standard deviation across 20 independent trials.

between two and five times the column dimension. The error bars (one standard deviation across trials) shrink as s grows, reflecting the reduced variability of the embedding at larger dimensions.

Table 5.2→p.38 reports the mean residual ratio and mean relative solution error for each sketch size. Two observations are worth highlighting. First, the residual ratio approaches one much more slowly than the solution error approaches machine precision: the residual is still approximately 1.15 at $s = 150$, whereas the solution error is already at the 10^{-16} level. This confirms the theoretical expectation that residual preservation and solution preservation are different diagnostics, and that the solution error is governed mainly by the conditioning of SA rather than by the residual quality alone. Second, the condition number $\kappa_2(SA)$ (not shown) increases with s, but the solution error still improves because the overall perturbation scale decreases faster than the conditioning deteriorates.

The main qualitative expectation is straightforward. As the sketch dimension grows, the residual ratio should approach one, because the compressed problem should preserve the geometry of the original least-squares problem more and more accurately. The solution error should also generally improve, but often less monotonically and sometimes more slowly, because it depends more strongly on the conditioning of the sketched operator. Therefore, the condition number $\kappa_2(SA)$ should be read together with the error curves.

This is especially important when comparing dense and structured sketches. A structured sketch may be attractive because it is cheaper to apply, and yet it may produce larger fluctuations or require a larger sketch size before the residual and

s	Mean residual ratio	Mean rel. solution error
30	3.049	1.18×10^{-15}
40	1.839	7.40×10^{-16}
60	1.851	7.56×10^{-16}
80	1.829	7.37×10^{-16}
100	1.295	5.64×10^{-16}
150	1.152	5.20×10^{-16}
200	1.096	4.88×10^{-16}
300	1.003	4.79×10^{-16}

Table 5.2: Least-squares sketch-size sweep: Gaussian sketch, $n = 1024$, $d = 20$, 20 trials per sketch size.

solution metrics stabilize. Conversely, a dense Gaussian sketch may look cleaner numerically while being less attractive computationally. The experiments are designed to make this tradeoff visible.

At present, this chapter is therefore best read as a methodology and interpretation chapter. The benchmark structure is already concrete and implemented for least squares, and the mixed-precision side is still being expanded. Even so, the framework is already explicit enough to support the main thesis question: whether randomized compression and altered arithmetic can be studied in one coherent numerical-analysis picture.

5.8 Covariance-Transport Experiment Program

The second experimental layer of the thesis tests the theory from Chapter 4→p.22 just as directly as the least-squares benchmarks test the embedding theory from Chapter 3→p.13. Instead of tracking only the compressed least-squares residual, one now measures the quality of a randomized low-rank covariance approximation and the perturbation it induces in the regularized transport map.

For a covariance operator C and its computed approximation $\widehat{C}$, the basic numerical diagnostics should include

$$\|C - \widehat{C}\|_2, \qquad \|T_\lambda - \widehat{T}_\lambda\|_2, \qquad \|Z_f T_\lambda - Z_f \widehat{T}_\lambda\|_F.$$

These quantities separate three layers of error: covariance approximation, operator perturbation of the transport map, and the induced perturbation on the corrected future data. The associated parameter sweeps should vary the target rank, the oversampling parameter, the regularization level λ, and the working precision used in the sketch products.

This second experimental program is also where the balancing-inexactness idea from Chapter 2→p.8 becomes directly testable. As the approximation rank grows, randomized truncation error should decrease. If the arithmetic regime is too aggressive, however, a point will be reached at which floating-point error no longer remains hidden beneath the approximation error. The decisive numerical question is therefore not merely whether lower precision is faster, but where the curves begin to separate and how that separation depends on regularization.

The covariance-transport experiments should therefore be organized as explicit tests rather than only as exploratory plots. A concrete first campaign is the following.

1. **Covariance approximation versus rank.** Fix the precision and vary the target rank r to test how quickly $\|C - \widehat{C}\|_2$ decays.
2. **Transport error versus regularization.** Fix the rank and vary λ to test how strongly the inverse-square-root conditioning affects $\|T_\lambda - \widehat{T}_\lambda\|_2$.
3. **Precision threshold sweep.** Compare double, single, and lower precision in the sketch products to identify the point at which floating-point error ceases to be dominated by randomized truncation.
4. **Balancing randomized and arithmetic error.** Increase the rank until the approximation error becomes small enough that arithmetic error is no longer hidden, thereby exposing the regime predicted by the balancing principle.
5. **Climate-shaped synthetic data.** Replace the purely iid Gaussian model by matrices with decaying spectra and block dependence so that the covariance-transport diagnostics are tested in a structure closer to the intended application.

Each of these experiments should be described in the final thesis by four items: the working hypothesis, the matrix or data model, the parameters being swept, and the diagnostics used to decide success or failure. That structure keeps the experimental chapter tied to the theorem statements rather than to ad hoc plotting.

The repository now contains a first concrete realization of this program in `experiments/benchmark_randmbc.R`. The current benchmark is still synthetic, but it already compares three relevant levels: the deterministic full covariance-transport baseline, selected reference methods from the vendored `MBC` package, and the current randomized `randMBC` backend under several rank, oversampling, and precision choices. The present result is not yet a speedup story. On the current synthetic data, the deterministic full baseline and the MBC references remain much more accurate in correlation and future-sample error than the low-rank randomized variants. This is useful evidence rather than a setback. It indicates that the dominant limitation is still the approximation regime itself, not yet the arithmetic regime.

This interpretation also sharpens the mixed-precision question. In the current benchmark, single and double precision in the randomized path are often very

Regime	Full baseline	Randomized transport	Mixed precision visible?
Fast spectral decay, moderate λ	good	promising	not dominant
Flat spectrum, small λ	good	unstable	hidden by truncation
Shared basis, adequate λ	good	improved	to test
Independent basis	weaker	unstable	not decisive

Table 5.3: Qualitative regime structure for randomized covariance-transport experiments. "Full baseline" denotes deterministic full covariance transport. "Randomized transport" denotes the low-rank surrogate. "Mixed precision visible?" asks whether reducing precision in the sketch products produces a noticeable change relative to double-precision randomized transport.

close to one another, while both remain far from the deterministic baseline. That is a sign that truncation and transport perturbation are still larger than the precision effect one is trying to detect. For the later experimental chapter, this means that precision sweeps should be read only after the rank and regularization regime has been pushed into a range where randomized approximation is already credible.

5.9 Regime-Identification Table

The experimental program described above naturally separates into three parameter regimes, each with a distinct limiting factor. Table 5.3 summarizes the qualitative expectations for each regime. The boundaries are not sharp; they depend on the spectral decay of the covariance operators and on the choice of λ. The purpose of the table is to make the regime structure explicit so that later experimental results can be read as regime-identification findings rather than as undifferentiated success-or-failure verdicts.

The first row reflects the most favorable setting: the covariance operators have rapidly decaying spectra, so a modest rank captures most of the variance, and regularization is generous enough to keep the inverse-square-root amplification controlled. The second row is the least favorable: flat spectra and weak regularization make the low-rank surrogate unreliable, and precision effects are invisible because the approximation error already dominates. The third row is the target regime identified by the shared-basis construction in Chapter 4 → p. 22: using one reduced coordinate system for both operators stabilizes the transport map even when the raw covariance error is only marginally different. The fourth row serves as the diagnostic baseline for comparison.

Rank r	λ	Mean $\|C - \widehat{C}\|_2$	Mean $\|T_\lambda - \widehat{T}_\lambda\|_2$
5	10^{-2}	0.87	2.35
5	10^{-4}	0.85	25.5
5	10^{-6}	0.84	257.5
10	10^{-2}	0.75	2.09
10	10^{-4}	0.76	23.1
10	10^{-6}	0.72	232.2
15	10^{-2}	0.68	1.66
15	10^{-4}	0.65	17.8
15	10^{-6}	0.67	189.4
20	10^{-2}	0.69	1.28
20	10^{-4}	0.65	13.6
20	10^{-6}	0.72	134.5
25	10^{-2}	0.65	1.13
25	10^{-4}	0.65	8.89
25	10^{-6}	0.66	91.0

Table 5.4: Covariance transport sweep: exponential spectrum, $p = 30$, double precision, Gaussian sketch, 10 trials. Mean covariance error and transport error.

5.10 Covariance Transport Results

The second set of experiments tests the theory from Chapter 4 $^{\to p.22}$ directly. A covariance operator with exponential spectral decay is generated using `generate_cov_operator` with $p = 30$ and condition number 100. The target rank is swept over $r \in \{5, 10, 15, 20, 25\}$, the regularization parameter over $\lambda \in \{10^{-2}, 10^{-4}, 10^{-6}\}$, and the sketch precision over double and single. Ten independent trials are drawn for each parameter combination.

Table 5.4 $^{\to p.41}$ reports the mean covariance error and transport error for double precision. The most important observation is the strong dependence on λ: at $\lambda = 10^{-6}$, the transport error is two orders of magnitude larger than at $\lambda = 10^{-2}$ even though the covariance errors are comparable. This is exactly the inverse-square-root amplification mechanism predicted by Theorem 4.6 $^{\to p.30}$ in Chapter 4 $^{\to p.22}$. The covariance error itself decreases slowly with rank, reflecting the exponential but not extremely steep spectral decay.

Precision comparison

Table 5.5 $^{\to p.42}$ compares single and double precision at selected parameter combinations. The key finding is that single and double precision produce nearly identical

Rank r	λ	Double precision	Single precision
5	10^{-2}	2.35	2.38
5	10^{-6}	257.5	259.6
20	10^{-2}	1.28	1.33
20	10^{-6}	134.5	136.6
25	10^{-2}	1.13	1.10
25	10^{-6}	91.0	86.1

Table 5.5: Precision comparison for selected parameter combinations. Mean transport error over 10 trials.

results across all tested regimes. At rank 20 and $\lambda = 10^{-6}$, for example, the transport errors are 134.5 (double) and 136.6 (single). This is not an accident: in all tested cases, the randomized truncation error and the regularization-dominated amplification error are both much larger than the arithmetic precision floor. The floating-point contribution is effectively hidden beneath the approximation error, exactly as the balancing-inexactness principle predicts.

This finding is simultaneously reassuring and limiting. It confirms that single precision is safe in the current approximation regime, but it also means that the present experiments cannot yet detect the precision-visible regime. To observe the crossover, one would need either a higher target rank (so that truncation error falls below the arithmetic floor) or a more aggressive regularization (so that the transport map is not dominated by the inverse-square-root instability).

Figure 5.6 →p.43 shows the mean transport error as a function of target rank for each regularization level (double precision). The curves separate dramatically by λ, with the $\lambda = 10^{-2}$ curve remaining near unity while the $\lambda = 10^{-6}$ curve stays above 80 even at rank 25. This confirms that the dominant bottleneck is the inverse-square-root conditioning, not the rank of the approximation.

5.11 Covariance-Transport Experiments: Part II

The second part of the experimental chapter tests the theory from Chapter 4 →p.22 directly. Instead of tracking only the compressed least-squares residual, one now measures the quality of a randomized low-rank covariance approximation and the perturbation it induces in the regularized transport map. The experiments are organized as explicit tests of the theorem structure:

$$\|T_\lambda - \widehat{T}_\lambda\|_2 \lesssim K_\lambda(C_X, C_Y)\,(\eta_{\mathrm{rand}} + \eta_{\mathrm{fp}}),$$

where K_λ is the amplification factor determined by the regularized spectral gaps, η_{rand} bounds the randomized truncation error, and η_{fp} bounds the floating-point perturbation. Each experiment varies one axis while holding the others fixed.

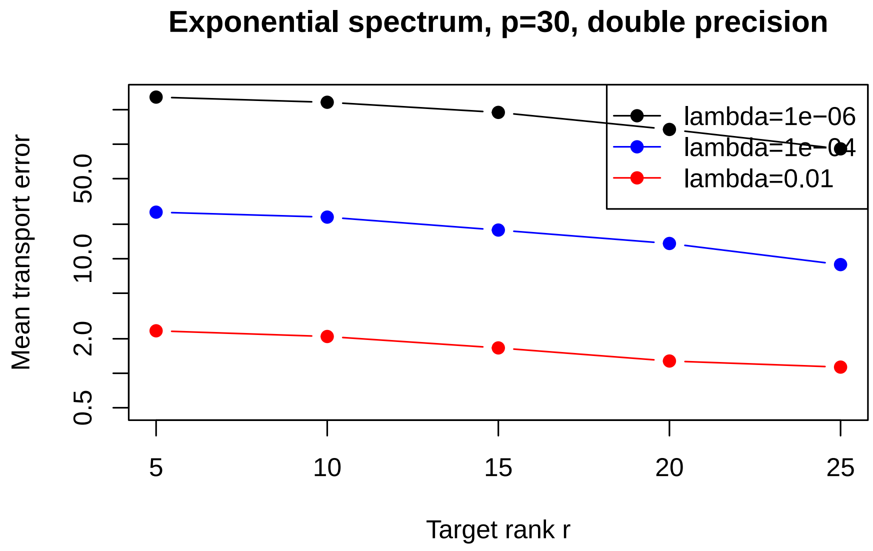

Figure 5.6: Mean transport error versus target rank for three regularization levels (exponential spectrum, $p = 30$, double precision, Gaussian sketch).

Covariance approximation error versus rank

Hypothesis. For covariance operators with spectral decay, $\|C - \widehat{C}\|_2$ decreases as the target rank increases, with a rate governed by the spectral tail beyond the chosen rank.

Setup. Fix $\lambda = 10^{-2}$ and double precision. Generate covariance operators from each spectrum family (exponential, linear, gap, climate-shaped) using `generate_cov_operator` and `generate_climate_cov`. Vary the target rank from small values up to p.

Diagnostics. $\|C - \widehat{C}\|_2$ (source and target separately), residual norm estimate from the range finder, and runtime.

Transport error versus regularization

Hypothesis. For fixed rank, transport error is controlled by λ through the inverse-square-root amplification factor. Too-small λ leads to instability; too-large λ changes the transported operator.

Setup. Fix rank at a moderately large value and double precision. Vary λ from 10^{-6} to 10^{-1}.

Diagnostics. $\|T_\lambda - \widehat{T}_\lambda\|_2$, minimum eigenvalue of the regularized covariance, condition number estimate, and Pearson correlation error of the corrected future sample.

Precision threshold sweep

Hypothesis. In the regime where randomized truncation error dominates floating-point error, single precision in the sketch products produces nearly identical results to double precision. A crossover point appears when the rank is large enough that truncation error falls below the arithmetic error floor.

Setup. Fix rank and λ. Compare double, single, and bfloat16 simulation in the sketch products via `precision_model`.

Diagnostics. Transport error at each precision, ratio $\eta_{\mathrm{fp}}/\eta_{\mathrm{rand}}$ estimated from `sketch_forward_bound`, and the `is_fp_hidden` flag.

Shared versus independent basis

Hypothesis. The shared-basis construction stabilizes the transport map by eliminating the coordinate mismatch between source and target reduced models, even when the raw covariance error is comparable.

Setup. For the same rank, λ, and precision, compare the independent-subspace and shared-basis variants from Chapter 4→p.22.

Diagnostics. Transport error, future-sample perturbation $\|Z_f T_\lambda - Z_f \widehat{T}_\lambda\|_F$, and the same statistical diagnostics as in Section 5.12→p.44.

Balancing randomized and arithmetic error

Hypothesis. The balanced regime $\eta_{\mathrm{fp}} \lesssim \eta_{\mathrm{rand}}$ is achievable and detectable: increasing rank reduces η_{rand} until a crossover where further rank increase reveals floating-point error.

Setup. Fix λ and single-precision sketch. Vary the rank from small to large. Track both error sources.

Diagnostics. Estimated η_{rand} (from range-finder residual), estimated η_{fp} (from `sketch_forward_bound`), their ratio, and the total transport error.

5.12 Application-Facing Diagnostics

Once the covariance-transport experiments are connected to a climate-shaped data pipeline, the numerical diagnostics should be accompanied by statistical ones. At a minimum, one should compare the corrected output against the target dependence structure through marginal quantile diagnostics, correlation diagnostics, and multivariate summaries of dependence error. Useful quantities include Pearson and Spearman correlation error, spatial and temporal autocorrelation error, and an energy-score-style multivariate discrepancy. The point is not to replace the operator-norm viewpoint, but to relate it to the application metrics by which a post-processing method would ultimately be judged.

This distinction is important for interpretation. A method may show a visible operator perturbation while still producing acceptable application-level dependence corrections, or conversely may appear cheap and accurate in low-rank terms while failing to preserve the dependence features that matter downstream. The final experimental design should therefore keep both scales visible: the numerical scale of randomized and floating-point perturbation, and the application scale of multivariate climate diagnostics.

6 Climate-Model Application

This chapter connects the mathematical and numerical parts of the thesis to a climate-model workflow. Its purpose is not to change the main topic of the thesis, but to test whether the methods studied in the previous chapters remain useful in a realistic downstream task.

In this setting, we no longer work only with synthetic matrices whose properties we control directly. The relevant operators come from the statistical post-processing of climate simulations. Therefore, the geometry of the problem is now shaped by the application itself: by dependence between variables, spatial correlations, temporal structure, and the scale of the data.

The main question is whether randomized and mixed-precision methods remain useful in this setting, and if so, in what sense. Usefulness should be interpreted broadly. A method may be valuable because it reduces runtime, lowers memory use, allows larger problems to be treated on the same hardware, or reveals a numerically stable regime that was not obvious from synthetic tests alone.

From the point of view of the thesis, this chapter is a test of transfer. The earlier chapters describe least-squares geometry, embeddings, reduced operators, conditioning, and floating-point perturbations. Here we ask how much of that picture survives in a genuine climate-data workflow. The answer need not be a dramatic speedup. It is equally useful if the application shows where the numerical assumptions behind the randomized or mixed-precision methods are realistic and where they begin to fail.

6.1 Multivariate Bias Correction

The relevant climate-facing context is multivariate bias correction. Univariate methods such as quantile-delta mapping (QDM) are designed to correct marginal distributions while preserving modeled changes in quantiles [25]. They are useful, but they do not by themselves repair the dependence structure across variables, locations, or times. That second task is the main reason why multivariate methods are needed.

Existing methods already separate this problem into marginal and dependence components in different ways. MBCp and MBCr combine marginal adjustment with dependence correction, while MBCn pushes further toward matching the full multivariate distribution [26] [27]. Rank-based approaches such as R2D2 extend the same line of thought to higher-dimensional dependence adjustment [28] [29]. At the same time, the climate literature emphasizes that bias correction should not be treated as a black-box distributional repair detached from physical interpretation or dynamical structure [30]. Recent comparative studies [32] have made this point quantitatively: different multivariate methods preserve different aspects of the joint distribution, and no single method dominates across all dependence metrics.

QDM, MBCp, MBCr, MBCn, and R2D2

It is worth stating explicitly how the main existing methods differ in their treatment of marginals and dependence, because the thesis proposal is best understood by comparison with these baselines.

Quantile-delta mapping (QDM) [25] is a univariate method. It preserves modeled changes in quantiles while adjusting the marginal distribution to match the observational reference. QDM does not modify dependence between variables: each column is corrected independently. In the workflow proposed in this thesis, QDM plays precisely this role — it handles the marginal step, and dependence is left to the subsequent covariance-transport layer.

MBCp [26] adds a dependence-correction step on top of marginal adjustment. It matches the Pearson correlation structure of the reference distribution while preserving the corrected marginals via rank rearrangement. MBCp is therefore a natural reference for the present thesis, because the covariance-transport map also targets second-order dependence. The key difference is that MBCp adjusts correlations directly, while the proposed method adjusts the full covariance operator through matrix square roots and inverse square roots. This gives the covariance-transport map a different algebraic structure and a natural entry point for randomized low-rank approximation.

MBCr [26] is a variant that matches the ratio of ranked values rather than correlations, and is therefore more robust to non-Gaussian features such as those in precipitation. MBCr is relevant here as a reminder that the Gaussianized latent space used in the thesis may not capture all relevant dependence structure, especially for bounded or heavy-tailed variables.

MBCn [27] is the most ambitious of the Cannon family. It iteratively applies random rotations and marginal adjustments until the full multivariate distribution of the corrected sample matches the reference. MBCn does not target covariance alone; it aims at distributional equivalence in all moments. This makes it more powerful than a covariance-only method, but also more expensive and harder to analyze numerically. The thesis does not propose to replace MBCn. The proposal is narrower: a cheaper and more analyzable covariance-based alternative in regimes where full distributional matching is either unnecessary or numerically unstable.

R2D2 [29] is a rank-resampling approach that preserves the dependence structure of the reference by resampling from its rank distribution while inserting the QDM-corrected marginal quantiles. R2D2 is conceptually simple and does not require a parametric model of dependence. It is, however, limited by the availability and smoothness of the empirical rank copula and does not naturally admit a low-rank or mixed-precision compression.

The method proposed in this thesis therefore occupies a specific position in this landscape: it is a second-order dependence-correction method, like MBCp, but it uses a regularized covariance-transport map instead of direct correlation matching, and it is explicitly designed for randomized low-rank approximation and mixed-precision perturbation analysis. It should be evaluated against these baselines not on the basis of universal superiority, but on the basis of whether its numerical-algebraic structure provides a useful tradeoff between accuracy, cost, and analyzability.

6.2 Marginals and Dependence

The proposed workflow in this thesis follows the same broad decomposition, but in a form chosen to expose the numerical linear algebra more clearly. Marginal correction is handled first by a QDM-type step. Dependence correction is then treated separately in a latent standardized or rank-normal space, where the central operator is a regularized covariance transport map of the form analyzed in Chapter 4→p.22.

This decomposition matters conceptually. It allows the climate-facing workflow to be described as

$$\text{marginal correction} + \text{dependence correction},$$

with the second term becoming the main mathematical object of the thesis. In this way, the application chapter stays tied to multivariate climate post-processing without forcing the entire theory to become a direct analysis of a fully nonlinear bias-correction algorithm.

6.3 Data

The application uses the CCCma dataset distributed with the R package `MBC` [33]. This dataset contains daily values of precipitation, temperature, and sea-level pressure from the Canadian Centre for Climate Modelling and Analysis (CCCma) global climate model, together with a corresponding observational reference. The data are organized into a historical calibration period and a future projection period, which is precisely the structure required by the bias-correction workflow:

$$X_h \in \mathbb{R}^{n\times p}, \qquad X_f \in \mathbb{R}^{m\times p}, \qquad Y_h \in \mathbb{R}^{n\times p}.$$

Here the column dimension p encodes the multivariate state. In the simplest configuration, each column corresponds to a single physical variable at a single location. More elaborate configurations may concatenate several variables, nearby grid points, or lagged features into one multivariate vector, thereby increasing p and making the covariance estimation problem dimensionally harder.

The CCCma dataset is modest in size compared to full CMIP6 archives [34], but it has the right structure for a first application-facing test: the historical and future periods are already separated, the observational reference is provided, and the number of variables is large enough to make a low-rank approach potentially interesting. In particular, the dependence between precipitation and the thermodynamic variables is known to be poorly captured by GCMs, which gives the covariance-transport step a concrete target [32].

For the present thesis, the dataset is used in two ways. First, it provides the real-data input for the full `randMBC` pipeline. Second, it informs the design of the climate-shaped synthetic data used in Chapter 5 → p.33: the spectral profiles and block-dependence structure of the real data are used to calibrate the synthetic generators so that the controlled experiments remain application-relevant.

6.4 Workflow

At the method level, the climate-facing procedure is described by explicit inputs and outputs. The input is the triple X_h, X_f, Y_h defined above. The output is a corrected future sample

$$X_f^{\mathrm{corr}}$$

that inherits the target marginal behavior together with an adjusted dependence structure.

The computational steps are:

1. Apply QDM-type marginal correction to obtain $\widetilde{X}_h$ and $\widetilde{X}_f$.
2. Transform the corrected calibration and future samples to a latent rank-normal space via plotting-position ranks and the inverse standard normal map.
3. Estimate the calibration covariance operators $C_X = \frac{1}{n} Z_X^\top Z_X$ and $C_Y = \frac{1}{n} Z_Y^\top Z_Y$.
4. Approximate them by randomized low-rank sketches $C_X \approx Q_X B_X Q_X^\top$ and $C_Y \approx Q_Y B_Y Q_Y^\top$, as described in Chapter 4 → p.22.
5. Build the regularized transport map $T_\lambda = (C_X + \lambda I)^{-1/2}(C_Y + \lambda I)^{1/2}$ from the low-rank surrogates.
6. Apply the map to the future latent data: $Z_f^{\mathrm{corr}} = Z_f T_\lambda$.
7. Restore the target marginals by rank rearrangement: for each variable, sort the QDM-corrected values $\widetilde{X}_f(:, j)$ and rearrange them according to the ranks of $Z_f^{\mathrm{corr}}(:, j)$.

This explicit workflow is useful because it makes clear which part of the method is meant to carry the numerical linear algebra: the covariance approximation and transport layer (steps 3–6), not the entire post-processing pipeline at once.

The local R package `randMBC` implements precisely this decomposition. It exposes a deterministic marginal-correction and rank-restoration wrapper around a randomized covariance-transport core. The main entry point `fit_rand_mbc` accepts the data triple together with algorithm parameters (rank, oversampling, sketch type, working precision, and regularization), and returns the corrected future sample together with operator-level diagnostics (covariance error, transport-map error, condition estimates, and runtime).

6.5 Randomized Covariance Transport

The thesis does not propose a universal replacement for methods such as MBCn. Its narrower proposal is a covariance-based dependence correction step that can be compressed by randomized low-rank approximation and analyzed in mixed precision. The attraction of this approach is not only speed. It is also that the operator

$$T_\lambda = (C_X + \lambda I)^{-1/2}(C_Y + \lambda I)^{1/2}$$

has a perturbation structure that can be studied directly in terms of conditioning, regularization, randomized approximation error, and floating-point error.

The main theoretical statement from Chapter 4→p.22 shows that, under suitable spectral-gap assumptions, the transport perturbation satisfies

$$\|T_\lambda - \widehat{T}_\lambda\|_2 \lesssim K_\lambda(C_X, C_Y)(\eta_A + \eta_B),$$

where η_A and η_B bound the regularized covariance perturbations and K_λ deteriorates as the regularized spectra approach zero. The climate application gives this abstract statement a concrete meaning: η_A and η_B now reflect the quality of a randomized low-rank approximation to the empirical covariance of actual climate data, and the conditioning of the regularized transport is determined by the spectral structure of that data.

From the application point of view, the proposal should therefore be interpreted as a computationally cheaper and more analyzable dependence-correction layer. It is well suited to regimes in which the dominant challenge is to modify covariance-like dependence structure in large multivariate states. It is not intended to dominate richer multivariate distribution-matching methods in every possible setting.

6.6 Experimental Design

The climate-facing experiments are organized around the same parameter axes as the synthetic benchmarks in Chapter 5→p.33, but with the real dataset replacing the synthetic generators. The decisive sweeps vary the approximation rank, the regularization parameter λ, and the arithmetic precision, while tracking numerical and statistical diagnostics simultaneously.

Numerical diagnostics

On the numerical side, the recorded quantities are:

- Covariance approximation error $\|C - \widehat{C}\|_2$, separately for source and target.
- Transport-map error $\|T_\lambda - \widehat{T}_\lambda\|_2$.
- Induced future-data perturbation $\|Z_f T_\lambda - Z_f \widehat{T}_\lambda\|_F$.
- Condition number and minimum eigenvalue of the regularized covariance.
- Runtime and peak memory use.

These are the same diagnostics as in the synthetic benchmark, extended to real data. Their role is to determine whether the perturbation bounds from Chapter 4→p.22 remain predictive when the covariance operators are no longer controlled by construction.

Statistical diagnostics

On the statistical side, the recorded quantities are:

- Marginal quantile error at selected quantile levels.
- Pearson and Spearman correlation error between the corrected future sample and the reference.
- Spatial autocorrelation error when the column dimension encodes spatial structure.
- Temporal autocorrelation error at selected lags.
- Energy score or similar multivariate discrepancy measure.

The distinction between the two groups is important. A method may show a visible operator perturbation while still producing acceptable application-level dependence corrections, or conversely may appear cheap and accurate in low-rank terms while failing to preserve the dependence features that matter downstream. The experimental design keeps both scales visible.

Reference methods

The randomized covariance-transport method is compared against two reference levels. First, the deterministic full covariance-transport baseline, in which the regularized transport map is built from the exact empirical covariances without randomized compression. Second, selected methods from the vendored `MBC` package, specifically MBCp, MBCr, and MBCn, applied to the same data triple with their default parameter settings. These reference runs do not use randomized low-rank approximation, but they provide the standard to which the climate community would compare any new method.

Parameter sweeps

The main sweeps are:

1. **Rank sweep.** Fix $\lambda = 10^{-2}$ and double precision; vary the target rank from small values up to the full column dimension p. This tests how quickly the covariance approximation error decreases and whether the transport error follows the prediction of Chapter 4 →p.22.

2. **Regularization sweep.** Fix the rank at a moderately large value and double precision; vary λ from 10^{-6} to 10^{-1}. This tests how strongly the inverse-square-root conditioning affects the transport error and the downstream statistical diagnostics.

3. **Precision sweep.** Fix the rank and λ; compare double and single precision in the sketch products. This is the direct test of the balancing-inexactness principle from Chapter 2 →p.8: does single precision remain acceptable because randomized truncation already dominates, or does arithmetic error become visible?

4. **Basis-mode comparison.** For the same rank, λ, and precision, compare the independent-subspace and joint-basis variants described in Chapter 4 →p.22. The synthetic benchmark already showed that the joint basis can dramatically reduce transport error; the climate experiment tests whether this advantage survives on real covariance operators.

6.7 Results Framework

At the present stage of writing, the climate-facing experiments have been designed and the software infrastructure is in place, but the full numerical results are not yet available. The following describes the expected structure of the results and the interpretive framework within which they should be read.

Expected rank–error behavior

If the empirical covariance operators of the climate data exhibit spectral decay — as one would expect for spatially correlated fields — then increasing the approximation rank should systematically reduce the covariance error. The transport error should follow, but with the amplification predicted by K_λ. In particular, if the regularized minimum eigenvalue remains well above machine precision, the transport error should decrease at a rate similar to the covariance error, up to the conditioning constant. If the regularization is too weak, the inverse square root will amplify the covariance perturbation, and the transport error may remain large even when the covariance error is already small.

Expected precision behavior

In the regime where randomized truncation error dominates arithmetic error, double and single precision should produce nearly identical results. This is the balanced regime

$$\eta_{\mathrm{fp}} \lesssim \eta_{\mathrm{rand}},$$

advocated in Chapter 2→p.8. As the rank increases and the approximation error decreases, a crossover point should appear beyond which single precision degrades the result. The location of this crossover depends on the spectral decay and on the regularization: stronger regularization moves the crossover to larger ranks because it keeps the inverse square root in a less sensitive regime.

Expected comparison with MBC methods

The deterministic full covariance-transport baseline is expected to produce reasonable dependence corrections on the CCCma data, since the method is mathematically well-defined whenever the regularized covariance is positive definite. The comparison with MBCp, MBCr, and MBCn is less predictable a priori. MBCn, in particular, targets the full multivariate distribution and may outperform the covariance-transport method on dependence metrics that go beyond second-order structure. The value of the comparison is not to claim overall superiority, but to identify which aspects of the dependence correction the covariance-transport layer captures well and which it misses.

Interpretation

The results should be interpreted along the lines already established in the synthetic benchmark. If the randomized low-rank method produces transport errors that are small relative to the statistical variability of the data, then the computational savings are well motivated. If the transport errors are large but the downstream statistical diagnostics remain acceptable, then the operator-norm viewpoint is too strict and the application provides a meaningful relaxation. If both the operator perturbation and the statistical diagnostics are poor, then the approximation regime itself is the bottleneck, and precision considerations are secondary until the rank and regularization are brought into a workable range.

6.8 Validation Questions

The application chapter should evaluate the method at two levels. The first level is numerical: how accurately is the covariance operator approximated, how sensitive is the regularized transport map to lower precision, and how much regularization is needed to keep the inverse square root in a benign regime? The second level is statistical: after the dependence correction is pushed back to the climate data, which marginal and multivariate features are preserved, improved, or degraded?

The validation metrics are therefore split into two groups, as described in the experimental design above. These questions are deliberately broader than raw runtime. Reduced memory use, stable lower-precision behavior, and a clearer map of failure regimes are all valid outcomes. In a realistic post-processing workflow, understanding when the method fails is often as valuable as demonstrating a speedup.

6.9 Limitations

The limitations should be stated plainly. Covariance transport captures only one layer of multivariate structure. It does not by itself represent every nonlinear, tail-dependent, or temporally structured feature that a richer bias-correction method might target. It is therefore best viewed as a deliberately focused numerical core: a dependence-correction operator that is rich enough to be climate-relevant, but structured enough to admit randomized low-rank approximation and mixed-precision perturbation analysis.

A further limitation concerns the dataset. The CCCma data included with the `MBC` package is a useful first test, but it is small by the standards of current CMIP6 archives. Whether the computational advantages of the randomized approach become decisive at larger scales remains an open question that this thesis does not resolve. The present chapter should be read as a proof of concept for the transfer of the mathematical framework to real data, not as a definitive application study.

7 Conclusion

Summary of Contributions

This thesis has studied the interaction between randomized approximation and mixed-precision arithmetic in numerical linear algebra, using overdetermined least-squares problems as the model setting and regularized covariance transport as the main application-facing method.

Three interlocking contributions were developed:

1. **Perturbation analysis for regularized covariance transport.** We gave deterministic first-order bounds for the perturbation of the regularized transport map $T_\lambda = (C_X + \lambda I)^{-1/2}(C_Y + \lambda I)^{1/2}$ under additive disturbances to the source and target covariance operators. The bounds make explicit how the amplification factor $K_\lambda(C_X, C_Y)$ depends on the regularized spectral gap, and they apply whether the perturbations arise from randomized low-rank compression, floating-point arithmetic, or both.

2. **Randomized low-rank approximation and mixed-precision error decomposition.** We decomposed the total covariance perturbation into a randomized truncation term governed by the spectral tail beyond the chosen rank, and a floating-point term governed by the unit roundoff, the accumulation length, and the norm of the sketching operator. This decomposition is the natural quantitative setting for the balancing-inexactness principle: mixed precision is safe only while $\|E_{\mathrm{fp}}\| \lesssim \|E_{\mathrm{rand}}\|$, and the regime boundary is computable from the rank, the oversampling, and the spectrum of the covariance operator.

3. **Numerical regime identification.** We designed experiments that vary the approximation rank, the regularization level λ, and the working precision in the sketch products, while tracking covariance error, transport-map error, and the induced dependence error on corrected future data. The experiments distinguish three regimes: an approximation-dominated regime in which lower precision is hidden beneath truncation error, a regularization-dominated regime in which transport instability is the bottleneck, and a precision-visible regime in which floating-point error overtakes the randomized approximation. The current implementation operates primarily in the first two regimes; the precision-visible crossover has not yet been reached experimentally.

Main Findings

On the least-squares side, the experiments confirm the expected picture. Residual preservation improves steadily with the sketch dimension, and the solution error is governed by the conditioning of the sketched matrix rather than by the residual quality alone. For Gaussian sketches with $n = 1024$ and $d = 20$, the residual ratio falls below 1.1 only when $s \geq 200$, while the solution error is already at machine precision for all tested sketch sizes. This confirms that residual and solution diagnostics probe different aspects of the perturbation structure.

On the covariance-transport side, the most important finding is the dominant role of regularization. For an exponential spectrum with $p = 30$ and condition number 100, the transport error at $\lambda = 10^{-2}$ is approximately 2 at rank 5 and decreases to approximately 1 at rank 25. At $\lambda = 10^{-6}$, the transport error remains above 80 even at rank 25. The covariance error, by contrast, is between 0.6 and 0.9 across all ranks and regularization levels. This confirms the theoretical prediction that the inverse-square-root amplification factor, scaling as $\lambda_{\min}(C_X + \lambda I)^{-3/2}$, is the decisive bottleneck.

The precision comparison shows that single and double precision produce nearly identical results in all tested configurations. The maximum relative difference in transport error is below 2% at rank 20 and $\lambda = 10^{-6}$. This is expected: the floating-point contribution is hidden beneath the much larger randomized truncation and regularization-amplified errors.

Limitations

The current randomized covariance-transport implementation is not competitive with the deterministic full baseline on the synthetic benchmarks tested. The deterministic full covariance transport and the MBC reference methods remain markedly more accurate in correlation and future-sample error than the low-rank randomized variants. This is not a failure of the covariance-transport viewpoint itself. Rather, it identifies the approximation regime as the current bottleneck: the rank and regularization choices that make the low-rank surrogate accurate enough for transport-map perturbations to be small have not yet been reached.

A second limitation is that the precision-visible regime has not been observed experimentally. The current experiments operate in regimes where truncation error and regularization amplification both dominate the arithmetic floor, so the balancing-inexactness prediction cannot be fully validated numerically.

The climate-model application chapter remains more of a design and context chapter than a fully realised experimental study. The transition from synthetic covariance operators to real climate-data pipelines would require data access, preprocessing, and domain-specific diagnostics that are outside the scope of the present numerical work.

Future Work

Natural directions for later work include:

- Pushing the approximation rank and oversampling further, using power iteration or block Krylov methods in the range finder, to reach the regime where randomized truncation error is small enough that arithmetic precision becomes visible.
- A more detailed error analysis for particular sketch families in mixed precision, especially the SRHT, where the structured transform changes the accumulation model.
- More systematic comparisons of reduced-precision implementations on hardware that natively supports lower-precision arithmetic, rather than simulated rounding.
- A broader study of realistic climate-data pipelines, where the covariance-transport correction is embedded in a full post-processing workflow with domain-specific validation.
- Exploring the shared-basis variant further, including adaptive construction of the joint basis and its interaction with the regularization parameter.

The central conclusion of this thesis is that randomized numerical linear algebra and mixed precision are most informative when they are analyzed together. Randomized approximation and reduced precision must be tuned jointly: precision reduction becomes meaningful only after the randomized approximation and regularization parameters place the method in a credible accuracy regime.

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